From the 2019 Administration

AP® Macroeconomics

Practice Exam

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Note: This publication shows the page numbers that appeared in the ***2018–19 AP Exam Instructions*** book and in the actual exam. This publication was not repaginated to begin with page 1.

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AP Macroeconomics Exam

Regularly Scheduled Exam Date: Wednesday afternoon, May 15, 2019

Late-Testing Exam Date: Friday afternoon, May 24, 2019

AP Microeconomics Exam

Regularly Scheduled Exam Date: Friday morning, May 17, 2019

Late-Testing Exam Date: Wednesday morning, May 22, 2019

Section I	**Total Time:** 1 hour and 10 minutes Calculator not permitted **Number of Questions:** 60 *(The number of questions may vary slightly depending on the form of the exam.)* **Percent of Total Score:** 66.67% **Writing Instrument:** Pencil required
Section II	**Total Time:** 1 hour *(10-minute reading period, 50-minute writing period)* Calculator not permitted **Number of Questions:** 3 required free-response questions **Percent of Total Score:** 33.33% **Writing Instrument:** Pen with black or dark blue ink

What Proctors Need to Bring to This Exam

- ☐ Exam packets
- ☐ Answer sheets
- ☐ AP Student Packs
- ☐ *2018-19 AP Coordinator's Manual*
- ☐ This book—*2018-19 AP Exam Instructions*
- ☐ AP Exam Seating Chart template
- ☐ School Code and Homeschool/Self-Study Codes
- ☐ Pencil sharpener
- ☐ Container for students' electronic devices (if needed)
- ☐ Extra No. 2 pencils with erasers
- ☐ Extra pens with black or dark blue ink
- ☐ Lined paper
- ☐ Stapler
- ☐ Watch
- ☐ Signs for the door to the testing room
  - "Exam in Progress"
  - "Phones of any kind are prohibited during the test administration, including breaks"

Macroeconomics

Before Distributing Exams: Check that the title on exam covers is ***Macroeconomics***, and is printed in **blue** on the Section I exam booklet covers. (On any large-type exams, the exam title will be printed in **black**.) If there are any exam booklets with a different title, contact the AP coordinator immediately.

Microeconomics

Before Distributing Exams: Check that the title on all exam covers is ***Microeconomics***, and is printed in **black**. If there are any exam booklets with a different title, contact the AP coordinator immediately.

SECTION I: Multiple Choice

› **Do not begin the exam instructions below until you have completed the appropriate General Instructions for your group.**

Make sure you begin the exam at the designated time. Remember, you must complete a seating chart for this exam. See pages 295–296 for a seating chart template and instructions. See the *2018-19 AP Coordinator's Manual* for exam seating requirements (pages 56–59).

Macroeconomics

If you are giving the regularly scheduled exam, say:
It is Wednesday afternoon, May 15, and you will be taking the AP Macroeconomics Exam. Look at your exam packet and confirm that the exam title is "AP Macroeconomics" and is printed in blue on the Section I booklet cover. [For large-type exams: **If you are taking a large-type exam, the exam title "AP Macroeconomics" is printed in black.] Raise your hand if your exam packet contains any title other than "AP Macroeconomics," and I will help you.**

If you are giving the alternate exam for late testing, say:
It is Friday afternoon, May 24, and you will be taking the AP Macroeconomics Exam. Look at your exam packet and confirm that the exam title is "AP Macroeconomics" and is printed in blue on the Section I booklet cover. [For large-type exams: **If you are taking a large-type exam, the exam title "AP Macroeconomics" is printed in black.] Raise your hand if your exam packet contains any title other than "AP Macroeconomics," and I will help you.**

Microeconomics

If you are giving the regularly scheduled exam, say:
It is Friday morning, May 17, and you will be taking the AP Microeconomics Exam. Look at your exam packet and confirm that the exam title is "AP Microeconomics" and is printed in black. Raise your hand if your exam packet contains any title other than "AP Microeconomics," and I will help you.

If you are giving the alternate exam for late testing, say:
It is Wednesday morning, May 22, and you will be taking the AP Microeconomics Exam. Look at your exam packet and confirm that the exam title is "AP Microeconomics" and is printed in black. Raise your hand if your exam packet contains any title other than "AP Microeconomics," and I will help you.

Once you confirm that all students have the correct exam, say:

In a moment, you will open the exam packet. By opening this packet, you agree to all of the AP Program's policies and procedures outlined in the *2018-19 Bulletin for AP Students and Parents*.

You may now remove the shrinkwrap from the outside only of your exam packet. Do not open the Section I booklet; do not remove the shrinkwrap from the Section II materials. Put the white seals and the shrinkwrapped Section II booklet aside. . . .

Carefully remove the AP Exam label found near the top left of your exam booklet cover. Place it on page 1 of your answer sheet on the light blue box near the top right corner that reads "AP Exam Label."

If students accidentally place the exam label in the space for the number label or vice versa, advise them to leave the labels in place. They should not try to remove the label; their exam can still be processed correctly.

Listen carefully to all my instructions. I will give you time to complete each step. Please look up after completing each step. Raise your hand if you have any questions.

Give students enough time to complete each step. Don't move on until all students are ready.

Read the statements on the front cover of the Section I booklet. . . .

Sign your name and write today's date. . . .

Now print your full legal name where indicated. . . .

Turn to the back cover of your exam booklet and read it completely. . . .

Give students a few minutes to read the entire cover.

Are there any questions? . . .

You will now take the multiple-choice portion of the exam. You should have in front of you the multiple-choice booklet and your answer sheet. You may never discuss the multiple-choice exam content at any time in any form with anyone, including your teacher and other students. If you disclose the multiple-choice exam content through any means, your AP Exam score will be canceled.

Open your answer sheet to page 2. You must complete the answer sheet using a No. 2 pencil only. Mark all of your responses beginning on page 2 of your answer sheet, one response per question. Completely fill in the circles. If you need to erase, do so carefully and completely. No credit will be given for anything written in the exam booklet. Scratch paper is not allowed, but you may use the margins or any blank space in the exam booklet for scratch work. Calculators are not allowed on any part of this exam. Are there any questions? . . .

You have 1 hour and 10 minutes for this section. Open your Section I booklet and begin.

Note Start Time _____ . Note Stop Time _____ .

Check that students are marking their answers in pencil on their answer sheets and that they have not opened their shrinkwrapped Section II booklets.

After one hour, say:

There are 10 minutes remaining.

After 10 minutes, say:

Stop working. Close your booklet and put your answer sheet on your desk, faceup. Make sure you have your AP number label and an AP Exam label on page 1 of your answer sheet. Sit quietly while I collect your answer sheets.

Collect an answer sheet from each student. Check that each answer sheet has an AP number label and an AP Exam label.

After all answer sheets have been collected, say:

Now you must seal your exam booklet using the white seals you set aside earlier. Remove the white seals from the backing and press one on each area of your exam booklet cover marked "PLACE SEAL HERE." Fold each seal over the back cover. When you have finished, place the booklet on your desk, faceup. I will now collect your Section I booklet. . . .

Collect a Section I booklet from each student. Check that each student has signed the front cover of the sealed Section I booklet.

There is a 10-minute break between Sections I and II. When all Section I materials have been collected and accounted for and you are ready for the break, say:

Please listen carefully to these instructions before we take a 10-minute break. All items you placed under your chair at the beginning of this exam, including your Student Pack, must stay there, and you are not permitted to open or access them in any way. Leave your shrinkwrapped Section II packet on your desk during the break. You are not allowed to consult teachers, other students, notes, textbooks, or any other resources during the break. You may not make phone calls, send text messages, check email, use a social networking site, or access any electronic or communication device. You may not leave the designated break area. Remember, you may never discuss the multiple-choice exam content with anyone, and if you disclose the content through any means, your AP Exam score will be canceled. Are there any questions? . . .

You may begin your break. Testing will resume at _____ .

SECTION II: Free Response

After the break, say:

May I have everyone's attention? Place your Student Pack on your desk. . . .

You may now remove the shrinkwrap from the Section II packet, but do not open the exam booklet until you are told to do so. . . .

Read the bulleted statements on the front cover of the exam booklet. Look up when you have finished. . . .

Now take an AP number label from your Student Pack and place it on the shaded box. If you don't have any AP number labels, write your AP number in the box. Look up when you have finished. . . .

Read the last statement. . . .

Using your pen, print the first, middle, and last initials of your legal name in the boxes and print today's date where indicated. This constitutes your signature and your agreement to the statements on the front cover. . . .

Now turn to the back cover. Using your pen, complete Items 1 through 3 under "Important Identification Information.". . .

Read Item 4. . . .

Are there any questions? . . .

If this is your last AP Exam, you may keep your Student Pack. Place it under your chair for now. Otherwise if you are taking any other AP Exams this year, leave your Student Pack on your desk and I will collect it now. . . .

Read the information on the back cover of the exam booklet. Do not open the booklet until you are told to do so. Look up when you have finished. . . .

Collect the Student Packs from students who are taking any other AP Exams this year.

Then say:

Are there any questions? . . .

The total Section II time is one hour. This includes a 10-minute reading period. The reading period is designed to provide you with time to develop thoughtful, well-organized responses. You are advised to take advantage of the reading period to plan what you will write. You may begin writing your exam responses before the reading period is over. You may make notes on page 3 and the pages that contain the exam questions, but your responses must be written on the designated lined pages. Are there any questions? . . .

You are responsible for pacing yourself and may proceed freely from one question to the next. You must write your answers in the Section II booklet using a pen with black or dark blue ink. You are not permitted to use other colored pens or pencils to draw graphs or diagrams. If you need more paper to complete your responses, raise your hand. At the top of each extra sheet of paper you use, write only:

- **your AP number,**
- **the exam title, and**
- **the question number you are working on.**

Do not write your name.

Make sure to begin your response on the lined page directly following the first appearance of the question and continue your response on the additional lined pages that follow. Remember to write your answers on the lined pages provided for each question; only responses written on the lined pages will be scored.

You may now open the Section II booklet and begin the 10-minute reading period.

Note Start Time _____ . Note Stop Time _____ .

After 10 minutes, say:

The reading period is over. You have 50 minutes remaining to complete Section II.

Note Start Time _____ . Note Stop Time _____ .

Check that students are using pens and that they are writing their answers in their exam booklets.

After 40 minutes, say:

There are 10 minutes remaining.

After 10 minutes, say:

Stop working and close your exam booklet. Place it on your desk, faceup. . . .

If any students used extra paper for a question in the free-response section, have those students staple the extra sheet(s) to the first page corresponding to that question in their free-response exam booklets. Complete an Incident Report after the exam and return these free-response booklets with the extra sheets attached in the Incident Report return envelope (see page 68 of the *2018-19 AP Coordinator's Manual* for complete details).

Then say:

Remain in your seat, without talking, while the exam materials are collected. . . .

Collect a Section II booklet from each student. Check for the following:

- Exam booklet front cover: The student placed an AP number label on the shaded box and printed their initials and today's date.
- Exam booklet back cover: The student completed the "Important Identification Information" area.

When all exam materials have been collected and accounted for, return to students any electronic devices you may have collected before the start of the exam.

If you are giving the regularly scheduled exam, say:

You may not discuss or share the free-response exam content with anyone unless it is released on the College Board website in about two days. Your AP Exam score results will be available online in July.

If you are giving the alternate exam for late testing, say:

None of the content in this exam may ever be discussed or shared in any way at any time. Your AP Exam score results will be available online in July.

If any students completed the AP number card at the beginning of this exam, say:

Please remember to take your AP number card with you. You will need the information on this card to view your scores and order AP score reporting services online.

Then say:

You are now dismissed.

After-Exam Tasks

Be sure to give the completed seating chart to the AP coordinator. Schools must retain seating charts for at least six months (unless the state or district requires that they be retained for a longer period of time). Schools should not return any seating charts in their exam shipments unless they are required as part of an Incident Report.

NOTE: If you administered exams to students with accommodations, review the *2018-19 AP Coordinator's Manual* and the *2018-19 AP SSD Guidelines* for information about completing the Nonstandard Administration Report (NAR) form, and returning these exams.

The exam proctor should complete the following tasks if asked to do so by the AP coordinator. Otherwise, the AP coordinator must complete these tasks:

- Complete an Incident Report for any students who used extra paper for the free-response section. (Incident Report forms are provided in the coordinator packets sent with the exam shipments.) **These forms must be completed with a No. 2 pencil.** It is best to complete a single Incident Report for multiple students per exam subject, per administration (regular or late testing), as long as all required information is provided. Include all exam booklets with extra sheets of paper in an Incident Report return envelope (see page 68 of the *2018-19 AP Coordinator's Manual* for complete details).
- Return all exam materials to secure storage until they are shipped back to the AP Program. (See page 27 of the *2018-19 AP Coordinator's Manual* for more information about secure storage.) Before storing materials, check the "School Use Only" section on page 1 of the answer sheet and:
  - Fill in the appropriate section number circle in order to access a separate AP Instructional Planning Report (for regularly scheduled exams only) or subject score roster at the class section or teacher level. See "Post-Exam Activities" in the *2018-19 AP Coordinator's Manual*.
  - Check your list of students who are eligible for fee reductions and fill in the appropriate circle on their registration answer sheets.

Name:_____

Answer Sheet for AP Macroeconomics

Practice Exam, Section I

No.	Answer
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AP® Macroeconomics Exam

SECTION I: Multiple Choice

2019

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

At a Glance

Total Time
1 hour and 10 minutes
Number of Questions
60
Percent of Total Score
66.67%
Writing Instrument
Pencil required
Electronic Device
None allowed

Instructions

Section I of this exam contains 60 multiple-choice questions. Fill in only the circles for numbers 1 through 60 on your answer sheet.

Indicate all of your answers to the multiple-choice questions on the answer sheet. No credit will be given for anything written in this exam booklet, but you may use the booklet for notes or scratch work. After you have decided which of the suggested answers is best, completely fill in the corresponding circle on the answer sheet. Give only one answer to each question. If you change an answer, be sure that the previous mark is erased completely. Here is a sample question and answer.

Sample Question	Sample Answer
Chicago is a (A) state (B) city (C) country (D) continent (E) village	Ⓐ ● Ⓒ Ⓓ Ⓔ

Use your time effectively, working as quickly as you can without losing accuracy. Do not spend too much time on any one question. Go on to other questions and come back to the ones you have not answered if you have time. It is not expected that everyone will know the answers to all of the multiple-choice questions.

Your total score on the multiple-choice section is based only on the number of questions answered correctly. Points are not deducted for incorrect answers or unanswered questions.

Form I
Form Code 4PBP4-S

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MACROECONOMICS

Section I

Time—1 hour and 10 minutes

60 Questions

Directions: Each of the questions or incomplete statements below is followed by five suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.

1. An economy is operating at a point inside its production possibilities curve (PPC). Which of the following will most likely cause the economy to move toward the current PPC in the short run?

   (A) A decrease in government spending
   (B) A decrease in inflation
   (C) An increase in human capital
   (D) An increase in employment
   (E) An increase in imports

2. An increase in which of the following will most likely promote economic growth?

   (A) Taxes on investment
   (B) The price level
   (C) Human capital
   (D) Consumption of nondurable goods
   (E) Interest rates

3. Which of the following will cause aggregate supply to increase in Country X?

   (A) An increase in personal income taxes
   (B) The discovery of low-cost alternative sources of energy
   (C) A decrease in labor productivity with no change in nominal wages
   (D) Depreciation of country X's currency on the foreign exchange market
   (E) An increase in the price level

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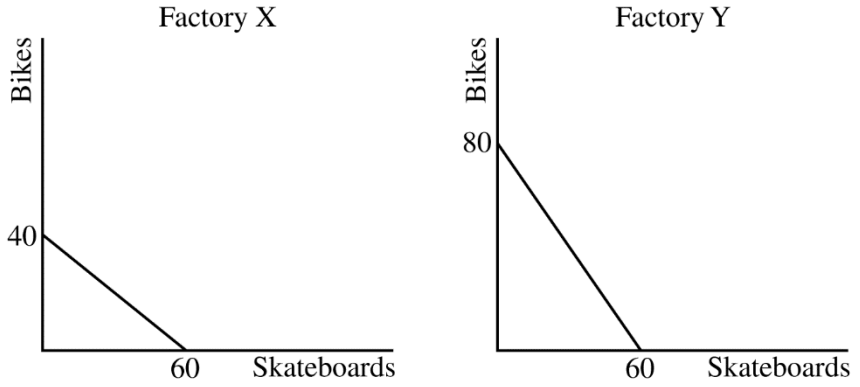

4. The graph above shows the production possibilities curve for Factory X and Factory Y. If Factory X uses the same amount of resources to produce skateboards and bikes as Factory Y uses, which of the following is true?

(A) Factory X has an absolute advantage in producing bikes.
(B) Factory X has an absolute advantage in producing skateboards.
(C) Factory X has a comparative advantage in producing skateboards.
(D) Factory Y has a comparative advantage in producing skateboards.
(E) Factory Y has an absolute advantage in producing skateboards.

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5. An increase in the price of oil, an important input to production, will result in which of the following in the short run?

(A) A decrease in the price level
(B) A decrease in short-run aggregate supply
(C) A decrease in unemployment
(D) An increase in real wages
(E) An increase in aggregate demand

6. An economy is currently operating at the full-employment level of output. Which of the following would result in a recessionary gap in the short run?

(A) An increase in the costs of production
(B) An improvement in the productivity of labor
(C) An increase in money supply
(D) A positive supply shock
(E) A decrease in income tax rates

7. Thailand and Malaysia are trading partners. If the price level in Thailand decreases relative to the price level in Malaysia, what will happen to Thailand's exports to Malaysia and Thailand's aggregate demand?

	Thailand's Exports	Thailand's Aggregate Demand
(A)	Increase	Decrease
(B)	Increase	Increase
(C)	Increase	Indeterminate
(D)	Decrease	Decrease
(E)	Decrease	Increase

8. Country X produces only apples and bananas. The following table shows prices and quantities of both products in two years.

	Year 1		Year 2	
	Price	Quantity	Price	Quantity
Apples	$1	100	$2	80
Bananas	$2	50	$2	60

Assuming year 1 is the base year, what is the nominal and real gross domestic product (GDP) for year 2 ?

	Nominal GDP	Real GDP
(A)	$200	$280
(B)	$200	$200
(C)	$280	$200
(D)	$280	$560
(E)	$560	$280

9. Which of the following is a fiscal policy action aimed at reducing unemployment?

(A) Decreasing government expenditures
(B) Decreasing income taxes
(C) Decreasing tax credits
(D) Increasing nominal interest rates
(E) Increasing required reserves

10. Increases in human capital can be achieved by which of the following?

(A) Building more factories
(B) Reducing immigration of skilled workers
(C) Improving the quality of job-training programs
(D) Increasing the physical capital per worker
(E) Increasing government spending on infrastructure

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11. If the marginal propensity to save is 0.25, a $15 billion increase in government spending will lead to an increase in national income by a maximum of

(A) $60 billion
(B) $45 billion
(C) $15 billion
(D) $11.25 billion
(E) $3.75 billion

12. Assume the economy is currently in long-run equilibrium. An increase in the money supply will affect unemployment in the short run and in the long run in which of the following ways?

	Short Run	Long Run
(A)	Falls	Falls
(B)	Rises	Rises
(C)	No change	Remains at the natural rate
(D)	Rises above the natural rate	Falls back to the natural rate
(E)	Falls below the natural rate	Rises back to the natural rate

13. The term "value added" for a firm is best defined as which of the following?

(A) The firm's sales
(B) The firm's sales minus depreciation
(C) The firm's sales minus its losses
(D) The firm's sales minus the cost of inputs purchased from other firms
(E) The firm's profits from its sales

14. An increase in a country's current account surplus will result in which of the following in the short run?

(A) A decrease in the country's government budget surplus
(B) A decrease in the country's national savings
(C) A decrease in the country's financial account deficit
(D) An increase in the country's net financial capital outflows
(E) An increase in the country's national debt

15. Using 2010 as the base year, the gross domestic product (GDP) deflator in 2011 was 97. Which of the following must be true?

(A) The inflation rate in 2011 was positive.
(B) The inflation rate in 2011 was negative.
(C) The inflation rate in 2011 was zero.
(D) The purchasing power of a dollar decreased by 3 percent.
(E) The real output increased by 3 percent.

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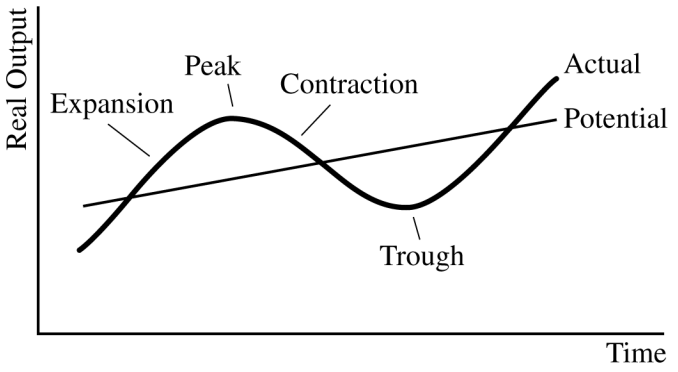

16. According to the business cycle represented in the diagram above, the actual rate of unemployment equals the natural rate of unemployment when the economy is

(A) in expansion
(B) in contraction
(C) at the peak
(D) at the trough
(E) on the potential line

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17. A fiscal policy action to reduce inflationary pressure would be to increase which of the following?

(A) The required reserve ratio
(B) The discount rate
(C) Transfer payments
(D) Government spending
(E) Income tax rates

18. The nominal gross domestic product of China in 2010 is a measure of the total value of which of the following in 2010 ?

(A) Financial assets, including stocks, in China
(B) Firms located within the borders of China
(C) Final goods and services consumed in China
(D) Final goods and services produced within the borders of China
(E) Final goods and services exported by China to the rest of the world

19. Assume the government reduces its spending and raises income taxes in an effort to reduce the budget deficit. The most likely short-run result will be an increase in

(A) interest rates
(B) unemployment
(C) the money supply
(D) the price level
(E) personal savings

20. Which of the following is true about inflation and interest rates?

(A) The higher the inflation rate, the higher the real interest rate.
(B) If there is no actual or expected inflation, the nominal and real interest rates are equal.
(C) If the economy is experiencing deflation, the nominal interest rate exceeds the real interest rate.
(D) The higher the inflation rate, the lower the nominal interest rate.
(E) The nominal interest rate is the difference between the real interest rate and the expected inflation rate.

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21. Economic growth is shown by a rightward shift in

(A) the aggregate demand curve
(B) the long-run Phillips curve
(C) the production possibilities curve
(D) the short-run aggregate supply curve
(E) the money supply curve

22. During a period of stagflation, a nation is most likely experiencing

(A) high unemployment and low inflation
(B) low unemployment and high inflation
(C) low inflationary expectations
(D) a decrease in short-run aggregate supply
(E) an increase in taxes and government spending

23. If businesses become optimistic about the profitability of investments in an economy, which of the following will happen in the loanable funds market in the short run?

(A) The supply and demand for loanable funds will increase.
(B) The supply and demand for loanable funds will decrease.
(C) The demand for loanable funds by the private sector will decrease.
(D) The real interest rate will increase.
(E) The real interest rate will decrease.

24. If investment demand becomes less responsive to changes in interest rates, which of the following is true?

(A) An expansionary fiscal policy results in less crowding out.
(B) An expansionary fiscal policy results in more crowding out.
(C) An expansionary monetary policy is more effective.
(D) A contractionary monetary policy is more effective.
(E) An expansionary monetary policy results in more crowding out.

25. Which of the following measures the opportunity cost of holding currency?

(A) The nominal wage rate
(B) The increase in the demand for money
(C) The forgone interest on alternative assets
(D) The ability to access currency to meet unexpected expenses
(E) The average income tax rates

26. A contractionary monetary policy combined with an expansionary fiscal policy will

(A) decrease both income and consumption
(B) increase both income and consumption
(C) have uncertain effects on the interest rate and investment
(D) increase the interest rate and decrease investment
(E) increase both the interest rate and investment

27. Ms. Smith withdraws $1,000 from her safe and deposits the money in a bank. If the bank holds no excess reserves and the reserve requirement is 10 percent, how will this deposit increase the bank's required reserves and the bank's loans?

	Required Reserves	Loans
(A)	$1,000	$9,000
(B)	$1,000	$10,000
(C)	$900	$100
(D)	$100	$900
(E)	$100	$1,000

28. A country can have an increased surplus in its balance of trade as a result of

(A) an increase in domestic inflation
(B) declining imports and rising exports
(C) higher tariffs imposed by its trading partners
(D) an increase in financial capital inflow
(E) an appreciating currency

29. Country X has a budget deficit. Which of the following changes in government budget outlays and tax revenues will result in a decrease in Country X's government budget deficit?

	Government Outlays	Tax Revenues
(A)	Fall by $100 million	Fall by $600 million
(B)	Fall by $200 million	Fall by $200 million
(C)	Rise by $300 million	Fall by $300 million
(D)	Rise by $400 million	Rise by $600 million
(E)	Rise by $500 million	Rise by $500 million

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GO ON TO THE NEXT PAGE.

	Motorcycles		Automobiles
Country X	50	OR	80
Country Y	75	OR	40

30. The table above shows the quantity of motorcycles and automobiles produced by two countries that use the same amount of resources. Which of the following is true?

(A) Country X has an absolute and comparative advantage in the production of motorcycles.
(B) Country X has an absolute and comparative advantage in the production of both goods.
(C) Neither country has a comparative advantage in the production of motorcycles.
(D) Country Y has an absolute and comparative advantage in the production of automobiles.
(E) Country Y has an absolute and comparative advantage in the production of motorcycles.

31. Which of the following is an example of how the consumer price index (CPI) exhibits bias in its estimates of changes in the cost of living?

(A) Energy prices have a higher impact on inflation than other input costs do.
(B) New products are always overrepresented in the CPI.
(C) The CPI assigns greater weight to measures of welfare than it does to economic activity.
(D) Product improvements are not always fully reflected in the calculation of the CPI.
(E) The CPI adjusts for the substitution of less expensive goods by consumers.

32. Fred Jones withdraws $1,000 in cash from his savings account. What immediate effect does this transaction have on the monetary aggregate measures of M1 and M2 ?

	M1	M2
(A)	Increases	Decreases
(B)	Increases	No change
(C)	Decreases	No change
(D)	No change	Decreases
(E)	No change	No change

33. Which of the following is true about inflationary expectations?

(A) The actual unemployment rate equals the natural rate of unemployment if the actual inflation rate exceeds the expected inflation rate.
(B) The actual unemployment rate equals the natural rate of unemployment when wages fully adjust to expected inflation.
(C) Expectations are always correct in the short run.
(D) The actual inflation rate is always equal to the expected inflation rate because of labor contracts.
(E) The natural rate of unemployment equals the inflation rate when the actual inflation rate equals the expected inflation rate.

34. Which of the following transactions is included in the financial account of Country X's balance of payments accounts?

(A) A firm in Country X sells robots to a firm in Country A.
(B) Country X sends financial aid to Country B.
(C) An individual in Country X receives dividend payments from a firm in Country C.
(D) An individual in Country X sends money monthly to family members in Country D.
(E) An individual in Country X buys new government bonds issued by Country E.

35. Assume a country's government increases taxes and its central bank decreases the money supply. The actions will result in an increase in which of the following in the short run?

(A) Aggregate demand
(B) Aggregate supply
(C) Investment spending
(D) Unemployment
(E) Inflation

36. When there is excess demand in the loanable funds market, which of the following will occur?

(A) National savings will exceed investment spending.
(B) The economy will remain at full employment.
(C) Real interest rates will increase.
(D) An inflationary gap will exist.
(E) The money supply will increase.

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37. Which of the following is an assumption underlying an upward-sloping short-run aggregate supply curve?

(A) The economy is experiencing high inflation.
(B) The economy is at full employment.
(C) National income is fixed.
(D) Wages are sticky.
(E) The velocity of money is constant.

38. The short-run Phillips curve implies there is a trade-off between

(A) rule making and discretionary policies
(B) monetary and fiscal policies
(C) inflation and unemployment
(D) budget deficits and interest rates
(E) interest rates and investment

39. When Stephanie took out a one-year fixed-rate loan, she expected to pay a real interest rate of 3 percent. At the end of the year, the real interest rate had fallen to 2 percent. Which of the following could have caused the decrease in the real interest rate?

(A) There was an increase in the nominal interest rate.
(B) There was a decrease in the nominal interest rate.
(C) There was a decrease in the money supply.
(D) The actual inflation rate was greater than the expected inflation rate.
(E) The actual inflation rate was less than the expected inflation rate.

40. A continuous increase in the consumer price index (CPI) is

(A) deflation
(B) stagflation
(C) inflation
(D) recession
(E) disinflation

41. Which of the following would most likely lead to cost-push inflation in the short run?

(A) A decrease in labor productivity
(B) A decrease in income tax rates
(C) A decrease in consumers' and businesses' optimism about future economic activity
(D) An increase in government deficit spending to stimulate a weak economic recovery
(E) Discovery of new sources of energy

42. The measured unemployment rate is often criticized for understating the level of joblessness because

(A) individuals working in the underground economy are counted as employed
(B) individuals working more than one job are counted more than once
(C) discouraged workers are counted as unemployed
(D) discouraged workers are not counted in the labor force
(E) part-time workers are counted as unemployed

43. An economy is in short-run equilibrium at a level of output that is greater than potential output. If there were no active fiscal or monetary policy intervention, which of the following changes in output and the price level would occur in the long run?

	Output	Price Level
(A)	Increase	Decrease
(B)	Increase	Increase
(C)	Decrease	Decrease
(D)	Decrease	Increase
(E)	No change	No change

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Production Point	Shirts	Chairs
A	20	0
B	16	1
C	12	2
D	8	3
E	4	4
F	0	5

44. The table shows the production possibilities for Country X in producing shirts and chairs when it uses all its available resources. The opportunity cost of producing one additional chair is

(A) zero
(B) constant
(C) increasing
(D) decreasing
(E) indeterminate

45. An increase in which of the following will most likely cause an increase in aggregate demand and inflation in the short run?

(A) Income tax rates
(B) Input prices
(C) Government spending
(D) Real interest rates
(E) Savings

46. If the central bank of a country wishes to maintain a stable nominal interest rate after a decrease in consumers' spending, taking which of the following actions will achieve the goal?

(A) Increasing government spending
(B) Increasing income tax rates
(C) Decreasing the required reserve ratio
(D) Decreasing the discount rate
(E) Selling government bonds

47. Investment in physical capital is most likely to occur as a result of an increase in

(A) interest rates
(B) inflation rates
(C) business confidence
(D) money demand
(E) personal consumption

48. Country X's government increases its spending without raising taxes. Which of the following is true about the effect on Country X's real interest rates and its subsequent effect on Country X's net exports?

(A) Real interest rates increase and net exports increase.
(B) Real interest rates increase and net exports decrease.
(C) Real interest rates increase with no change in net exports.
(D) Real interest rates decrease and net exports increase.
(E) Real interest rates decrease and net exports decrease.

49. Which of the following shifts the money demand curve to the right?

(A) An increase in the price level
(B) A decrease in the price level
(C) An increase in interest rates
(D) A decrease in interest rates
(E) A decrease in the nominal gross domestic product

50. An increase in the money supply will result in an increase in

(A) output in the short run and in the long run
(B) inflation in the short run and an increase in output in the long run
(C) inflation in the short run and no change in output in the long run
(D) output in the long run but not in the short run
(E) output in the long run and no change in inflation in the short run

51. Which of the following is an example of frictional unemployment?

(A) A former mayor doing volunteer work
(B) A factory worker who loses her job because of recession
(C) A college student working part-time at the campus bookstore
(D) A college graduate interviewing for two available positions
(E) An architect whose job is replaced by computer software that designs buildings

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52. Assuming no government policies, which of the following will occur in the long run if the actual unemployment rate exceeds the natural rate of unemployment?

(A) Prices will increase.
(B) Unemployment will increase.
(C) Wages will fall.
(D) Aggregate demand will increase.
(E) Long-run aggregate supply will decrease.

53. Assume that the market for bottled water is in equilibrium. If both the supply of and the demand for bottled water decrease, what will be the effect on equilibrium price and quantity?

	Price	Quantity
(A)	Decrease	Decrease
(B)	Decrease	Increase
(C)	Increase	Decrease
(D)	Increase	Indeterminate
(E)	Indeterminate	Decrease

54. An increase in government spending that is financed by an equal increase in taxes results in which of the following changes in aggregate demand (AD) and short-run aggregate supply (SRAS) curves?

	AD Curve	SRAS Curve
(A)	Shifts to the right	Shifts to the right
(B)	Shifts to the left	Shifts to the left
(C)	Shifts to the right	No change
(D)	Shifts to the left	No change
(E)	No change	No change

55. Assume that the United States current account balance is zero. If the United States dollar appreciates against the Japanese yen, then demand for United States exports will

(A) increase and result in a deficit in the United States financial account
(B) increase and result in a surplus in the United States financial account
(C) decrease and result in a surplus in the United States financial account
(D) decrease and result in a deficit in the United States financial account
(E) remain unchanged because the surplus in the current account will be offset by the deficit in the financial account

56. If the government decreases spending while the country's central bank buys bonds on the open market, which of the following will definitely occur?

(A) The aggregate demand curve will shift to the right.
(B) The aggregate demand curve will shift to the left.
(C) The short-run aggregate supply curve will shift to the right.
(D) Interest rates will fall.
(E) Interest rates will rise.

57. Which of the following best illustrates rising productivity?

(A) An expansion of the labor force
(B) An increase in the value of financial capital
(C) A decrease in the amount of physical capital per worker
(D) A decrease in the amount of labor needed to produce a unit of output
(E) An increase in the amount of resources required to produce a certain level of output

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58. Which of the following will most likely cause a depreciation in a country's currency?

(A) An increase in the country's price level
(B) An increase in the country's real interest rate
(C) A decrease in the country's expected inflation
(D) A decrease in the country's real gross domestic product
(E) A decrease in the country's money supply

59. Banks create money when

(A) they make loans
(B) the loans they make are repaid
(C) they keep all excess reserves
(D) customers increase their cash withdrawals from their savings accounts
(E) the money multiplier is less than one

60. Which of the following is true about a country's national debt?

(A) It is the sum of the country's trade deficit and government budget deficit.
(B) It increases when gross domestic product increases.
(C) It increases when the country's government has a budget deficit.
(D) It decreases when the country's exports exceed its imports.
(E) It decreases when national savings decrease.

END OF SECTION I

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION.

DO NOT GO ON TO SECTION II UNTIL YOU ARE TOLD TO DO SO.

MAKE SURE YOU HAVE DONE THE FOLLOWING.

- **PLACED YOUR AP NUMBER LABEL ON YOUR ANSWER SHEET**
- **WRITTEN AND GRIDDED YOUR AP NUMBER CORRECTLY ON YOUR ANSWER SHEET**
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AP® Macroeconomics Exam

SECTION II: Free Response

2019

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

At a Glance

Total Time
1 hour
Number of Questions
3
Percent of Total Score
33.33%
Writing Instrument
Pen with black or dark blue ink
Electronic Device
None allowed

Reading Period

Time
10 minutes. Use this time to read the questions and plan your answers. You may begin writing your responses before the reading period is over.

Writing Period

Time
50 minutes

Question 1
Suggested Time
25 minutes
Percent of Section II Score
50%

Question 2
Suggested Time
12.5 minutes
Percent of Section II Score
25%

Question 3
Suggested Time
12.5 minutes
Percent of Section II Score
25%

IMPORTANT Identification Information

PLEASE PRINT WITH PEN:

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   First letter of your first name ☐
2. Date of birth
   ☐☐ Month ☐☐ Day ☐☐☐☐ Year
3. Six-digit school code
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Instructions

The questions for Section II are printed in this booklet. You may use page 3 and the pages the questions are printed on to organize your answers and for scratch work, but you must write your answers on the lined pages provided for each question.

The proctor will announce the beginning and end of the reading period. You are advised to spend the 10-minute period reading all the questions and planning your answers. You may begin writing your responses before the reading period is over.

Write clearly and legibly. Do not skip lines. Cross out any errors you make; crossed-out work will not be scored.

Manage your time carefully. You may proceed freely from one question to the next. You may review your responses if you finish before the end of the exam is announced.

Form I
Form Code 4PBP4-S

MACROECONOMICS

Section II

Total Time—1 hour

Reading Period—10 minutes

Writing Period—50 minutes

Directions: You are advised to spend the first 10 minutes reading all of the questions and planning your answers. You will then have 50 minutes to answer all three of the following questions. You may begin writing your responses before the reading period is over. It is suggested that you spend approximately half your time on the first question and divide the remaining time equally between the next two questions. Include correctly labeled diagrams, if useful or required, in explaining your answers. A correctly labeled diagram must have all axes and curves clearly labeled and must show directional changes. Use a pen with black or dark blue ink.

Question 1 begins on page 4.

Question 2 begins on page 10.

Question 3 begins on page 14.

THIS PAGE MAY BE USED FOR TAKING NOTES AND PLANNING YOUR ANSWERS.

NOTES WRITTEN ON THIS PAGE WILL NOT BE SCORED.

WRITE ALL YOUR RESPONSES ON THE LINED PAGES.

1. Assume that a country's economy is currently in recession.

   (a) Draw a correctly labeled graph of the long-run and short-run Phillips curves. Label the current equilibrium as point Z.

   (b) Assume banks in the country hold no excess reserves and the public's holding of currency is constant. The required reserve ratio is 25%. The central bank of the country buys $100 billion in bonds from the nonbank public.

      (i) By how much will the monetary base of the country change?

      (ii) Calculate the change in the amount of loans in the banking system in the country.

      (iii) Calculate the change in the money supply in the country.

   (c) Draw a correctly labeled graph of the money market and show the effect of the change in the money supply identified in part (b)(iii) on the nominal interest rate.

   (d) Assume there is no change in inflationary expectations. On your graph in part (a), label a point W that is consistent with the effect of the change in the nominal interest rate identified in part (c).

   (e) Based on the change in the interest rate identified in part (c), how will the international value of the country's currency change?

   (f) Based on your answer in part (e), will the country's net exports increase, decrease, or stay the same? Explain.

THIS PAGE MAY BE USED FOR TAKING NOTES AND PLANNING YOUR ANSWERS.

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GO ON TO THE NEXT PAGE.

ANSWER PAGE FOR QUESTION 1

GO ON TO THE NEXT PAGE.

Question 1 is reprinted for your convenience.

1. Assume that a country's economy is currently in recession.

   (a) Draw a correctly labeled graph of the long-run and short-run Phillips curves. Label the current equilibrium as point Z.

   (b) Assume banks in the country hold no excess reserves and the public's holding of currency is constant. The required reserve ratio is 25%. The central bank of the country buys $100 billion in bonds from the nonbank public.

      (i) By how much will the monetary base of the country change?

      (ii) Calculate the change in the amount of loans in the banking system in the country.

      (iii) Calculate the change in the money supply in the country.

   (c) Draw a correctly labeled graph of the money market and show the effect of the change in the money supply identified in part (b)(iii) on the nominal interest rate.

   (d) Assume there is no change in inflationary expectations. On your graph in part (a), label a point W that is consistent with the effect of the change in the nominal interest rate identified in part (c).

   (e) Based on the change in the interest rate identified in part (c), how will the international value of the country's currency change?

   (f) Based on your answer in part (e), will the country's net exports increase, decrease, or stay the same? Explain.

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ADDITIONAL PAGE FOR ANSWERING QUESTION 1

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ADDITIONAL PAGE FOR ANSWERING QUESTION 1

GO ON TO THE NEXT PAGE.

ADDITIONAL PAGE FOR ANSWERING QUESTION 1

GO ON TO THE NEXT PAGE.

2. Suppose that the United States government implements a fiscal policy that increases the budget surplus.

   (a) Draw a correctly labeled graph of the loanable funds market and show the effect of the increase in the budget surplus on the equilibrium real interest rate.

   (b) The European Union is a major trading partner of the United States. Given your answer in part (a) about the real interest rate, will the United States dollar appreciate or depreciate against the euro? Explain.

   (c) Suppose that the Federal Reserve, the central bank of the United States, decides to offset the change in the value of the dollar identified in part (b).

      (i) Would the Federal Reserve buy or sell the euro?

      (ii) Would the Federal Reserve buy or sell the dollar?

   (d) Suppose that the Federal Reserve wants to counteract the real interest rate change identified in part (a). What open-market operation would the Federal Reserve use?

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ANSWER PAGE FOR QUESTION 2

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ADDITIONAL PAGE FOR ANSWERING QUESTION 2

GO ON TO THE NEXT PAGE.

ADDITIONAL PAGE FOR ANSWERING QUESTION 2

GO ON TO THE NEXT PAGE.

Time	Price of Apples	Quantity of Apples	Price of Oranges	Quantity of Oranges
Year 1	4	15	10	4
Year 2	6	10	5	20

3. The table above shows data for the economy of Fruitland, which produces two goods, apples and oranges.
   (a) Calculate the nominal gross domestic product (GDP) for year 2.
   (b) Using year 1 as the base year, calculate real GDP for year 2.
   (c) Calculate the GDP deflator for year 2.
   (d) Assuming the market basket is composed of the quantities in year 1, calculate the consumer price index (CPI) for year 2.
   (e) Suppose apple and orange pickers received a 2 percent increase in their wages. Based on your answer to part (d), would real wages of apple and orange pickers increase, decrease, or stay the same from year 1 to year 2 ? Explain.

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ANSWER PAGE FOR QUESTION 3

GO ON TO THE NEXT PAGE.

ADDITIONAL PAGE FOR ANSWERING QUESTION 3

GO ON TO THE NEXT PAGE.

ADDITIONAL PAGE FOR ANSWERING QUESTION 3

GO ON TO THE NEXT PAGE.

STOP

END OF EXAM

THE FOLLOWING INSTRUCTIONS APPLY TO THE COVERS OF THE SECTION II BOOKLET.

- **MAKE SURE YOU HAVE COMPLETED THE IDENTIFICATION INFORMATION AS REQUESTED ON THE FRONT AND BACK COVERS OF THE SECTION II BOOKLET.**
- **CHECK TO SEE THAT YOUR AP NUMBER LABEL APPEARS IN THE BOX ON THE FRONT COVER.**
- **MAKE SURE YOU HAVE USED THE SAME SET OF AP NUMBER LABELS ON ALL AP EXAMS YOU HAVE TAKEN THIS YEAR.**

Answer Key for AP Macreconomics
Practice Exam, Section I

Question 1: D
Question 2: C
Question 3: B
Question 4: C
Question 5: B
Question 6: A
Question 7: B
Question 8: C
Question 9: B
Question 10: C
Question 11: A
Question 12: E
Question 13: D
Question 14: D
Question 15: B
Question 16: E
Question 17: E
Question 18: D
Question 19: B
Question 20: B
Question 21: C
Question 22: D
Question 23: D
Question 24: A
Question 25: C
Question 26: D
Question 27: D
Question 28: B
Question 29: D
Question 30: E
Question 31: D
Question 32: B
Question 33: B
Question 34: E
Question 35: D
Question 36: C
Question 37: D
Question 38: C
Question 39: D
Question 40: C
Question 41: A
Question 42: D
Question 43: D
Question 44: B
Question 45: C
Question 46: E
Question 47: C
Question 48: B
Question 49: A
Question 50: C
Question 51: D
Question 52: C
Question 53: E
Question 54: C
Question 55: C
Question 56: D
Question 57: D
Question 58: A
Question 59: A
Question 60: C

Multiple-Choice Section for Macroeconomics
2019 Course Framework Alignment and Rationales

Question 1

Skill	Learning Objective	Topic
2.A	MOD-1.B	Opportunity Cost and the Production Possibilities Curve (PPC)

(A)	Incorrect. A point inside the production possibilities curve means that the economy is operating below full employment. A decrease in government spending will decrease aggregate demand and reduce real output, not increase production to the maximum potential capacity as indicated by the PPC.
(B)	Incorrect. A point inside the production possibilities curve means that the economy is operating below full employment. A decrease in inflation would increase real wages and will decrease short-run aggregate supply and reduce real output, not increase production to the maximum PPC potential capacity as indicated by the PPC.
(C)	Incorrect. A point inside the production possibilities curve means that the economy is operating below full employment. An increase in human capital represents an increase in productivity, resulting in an increase in the maximum potential capacity the economy can produce and, therefore, an outward shift of the , rather than moving toward the current PPC.
(D)	**Correct**. A point inside the production possibilities curve means that the economy is operating below full employment. An increase in employment will therefore cause the economy to move toward the current PPC in the short run.
(E)	Incorrect. A point inside the production possibilities curve means that the economy is operating below full employment. An increase in imports will decrease aggregate demand and reduce real output, not increase production to the maximum potential capacity as indicated by the PPC.

Question 2

Skill	Learning Objective	Topic
2.A	MEA-2.B	Economic Growth

(A)	Incorrect. An increase in taxes on investment will reduce the rate of physical capital accumulation and will slow rather than promote economic growth in the long run.
(B)	Incorrect. Changes in the price level have no effect on economic growth.
(C)	**Correct**. An increase in human capital increases labor productivity, which increases output per capita and aggregate production in the long run.
(D)	Incorrect. An increase in the consumption of nondurable goods will decrease private savings and increase interest rates, reducing funding for private investment in plant and equipment, which reduces productivity and the accumulation of physical capital in the long run, thereby discouraging rather than promoting economic growth.
(E)	Incorrect. An increase in interest rates reduces funding for private investment in plant and equipment, which reduces productivity and the accumulation of physical capital in the long run, thereby discouraging rather than promoting economic growth.

Question 3

Skill	Learning Objective	Topic
2.A	MOD-2.C	Short-Run Aggregate Supply (SRAS)

(A)	Incorrect. An increase in personal income taxes will decrease private savings and increase interest rates, reducing funding for private investment in plant and equipment, which reduces productivity and accumulation of physical capital in the long run and decreases long-run aggregate supply.
(B)	**Correct**. Production costs are a determinant of aggregate supply. The discovery of low-cost alternative sources of energy will decrease production costs and increase aggregate supply.
(C)	Incorrect. A decrease in labor productivity with no change in nominal wages will decrease (not increase) short-run aggregate supply.
(D)	Incorrect. A depreciation of Country X's currency causes an increase in Country X's net exports and increases aggregate demand in Country X, not aggregate supply.
(E)	Incorrect. An increase in the price level causes a movement along the aggregate supply curve, not a rightward shift of the aggregate supply curve.

Question 4

Skill	Learning Objective	Topic
2.C	MKT-1.A	Comparative Advantage and Gains from Trade

(A)	Incorrect. Absolute advantage describes a situation in which one factory, using the same resources, can produce more of one good than another factory can. Factory X does not have an absolute advantage over Factory Y in producing bikes because Factory X can produce 40 bikes while Factory Y can produce 80 bikes. Therefore, Factory Y has an absolute advantage in producing bikes.
(B)	Incorrect. Absolute advantage describes a situation in which one factory, using the same resources, can produce more of one good than another factory can. Factory X does not have an absolute advantage over Factory Y in producing skateboards because both factories can produce 60 skateboards. Therefore, neither factory has an absolute advantage in producing skateboards.
(C)	**Correct**. Comparative advantage describes a situation in which a factory can produce a good at a lower opportunity cost than another factory. Factory X's opportunity cost of producing 1 skateboard is 0.67 of a bike, while Factory Y's opportunity cost of producing 1 skateboard is 1.33 bikes. Therefore, Factory X has a comparative advantage in producing skateboards.
(D)	Incorrect. Comparative advantage describes a situation in which a factory can produce a good at a lower opportunity cost than another factory. Factory Y's opportunity cost of producing 1 skateboard is 1.33 bikes while Factory X's opportunity cost of producing 1 skateboard is 0.67 of a bike. Therefore, Factory X (not Factory Y) has a comparative advantage in producing skateboards.
(E)	Incorrect. Absolute advantage describes a situation in which one factory, using the same resources, can produce more of one good than another factory can. Factory Y does not have an absolute advantage over Factory X in producing skateboards because both factories can produce 60 skateboards. Therefore, neither factory has an absolute advantage in producing skateboards.

Question 5

Skill	Learning Objective	Topic
3.A	MOD-2.C	Short-Run Aggregate Supply (SRAS)

(A)	Incorrect. An increase in the price of oil will increase production costs and decrease short-run aggregate supply, which causes an increase (not a decrease) in the price level.
(B)	**Correct**. Production costs are a determinant of short-run aggregate supply. An increase in the price of oil will increase production costs and decrease short-run aggregate supply.
(C)	Incorrect. An increase in the price of oil will increase production costs and decrease short-run aggregate supply, which causes an increase in the price level, a decrease in real output, and an increase (not a decrease) in unemployment.
(D)	Incorrect. An increase in the price of oil will increase production costs and decrease short-run aggregate supply, which causes an increase in the price level and a decrease (not an increase) in real wages.
(E)	Incorrect. An increase in the price of oil will increase production costs and decrease short-run aggregate supply (not increase aggregate demand).

Question 6

Skill	Learning Objective	Topic
2.A	MOD-2.H	Changes in the AD–AS Model in the Short Run

(A)	**Correct**. An increase in the cost of production causes the short-run aggregate supply curve to shift to the left, reducing real output below full employment. This will result in a recessionary gap.
(B)	Incorrect. An increase in the productivity of labor causes the short-run aggregate supply curve to shift to the right in the short run, increasing real output beyond full employment. This will result in an inflationary gap, not a recessionary gap.
(C)	Incorrect. An increase in the money supply causes the aggregate demand curve to shift to the right, increasing real output beyond full employment. This will result in an inflationary gap, not a recessionary gap.
(D)	Incorrect. A positive supply shock causes the short-run aggregate supply curve to shift to the right, increasing real output beyond full employment. This will result in an inflationary gap, not a recessionary gap.
(E)	Incorrect. A decrease in income tax rates causes the short-run aggregate supply curve and the aggregate demand curve to shift to the right in the short run, increasing real output beyond full employment. This will result in an inflationary gap, not a recessionary gap.

Question 7

Skill	Learning Objective	Topic
3.A	MOD-2.A	Aggregate Demand (AD)

(A)	Incorrect. When the price level in Thailand decreases relative to the price level in Malaysia, this will make domestic goods less expensive relative to foreign goods, which increases the Malaysian demand for Thai goods, increasing Thailand's exports and decreasing Thailand's imports. Therefore, Thailand's net exports increase and Thailand's aggregate demand increases (not decreases).
(B)	**Correct**. When the price level in Thailand decreases relative to the price level in Malaysia, this will make domestic goods less expensive relative to foreign goods, which increases the Malaysian demand for Thai goods, increasing Thailand's exports and decreasing Thailand's imports. Therefore, Thailand's net exports increase and Thailand's aggregate demand increases.
(C)	Incorrect. When the price level in Thailand decreases relative to the price level in Malaysia, this will make domestic goods less expensive relative to foreign goods, which increases the Malaysian demand for Thai goods, increasing Thailand's exports and decreasing Thailand's imports. Therefore, Thailand's net exports increase and Thailand's aggregate demand increases (is not indeterminate).
(D)	Incorrect. When the price level in Thailand decreases relative to the price level in Malaysia, this will make domestic goods less expensive relative to foreign goods, which increases the Malaysian demand for Thai goods, increasing (not decreasing) Thailand's exports and decreasing Thailand's imports. Therefore, Thailand's net exports increase and Thailand's aggregate demand increases (not decreases).
(E)	Incorrect. When the price level in Thailand decreases relative to the price level in Malaysia, this will make domestic goods less expensive relative to foreign goods, which increases the Malaysian demand for Thai goods, increasing (not decreasing) Thailand's exports and decreasing Thailand's imports. Therefore, Thailand's net exports increase and Thailand's aggregate demand increases.

Question 8

Skill	Learning Objective	Topic
1.C	MEA-1.J	Real v. Nominal GDP

(A)	Incorrect. This answer mixes the formulas for calculating the nominal and real GDP. Nominal GDP is calculated using Year 2 prices and quantities $(\$2 \times 80) + (\$2 \times 60) = \$280$, and the real GDP is calculated using Year 1 prices and Year 2 quantities $(\$1 \times 80) + (\$2 \times 60) = \$200$.
(B)	Incorrect. This answer uses Year 2 prices and quantities to calculate the nominal and real GDP. Nominal GDP is calculated using Year 2 prices and quantities $(\$2 \times 80) + (\$2 \times 60) = \$280$, and the real GDP is calculated using Year 1 prices and Year 2 quantities $(\$1 \times 80) + (\$2 \times 60) = \$200$.
(C)	**Correct**. Nominal GDP is calculated using Year 2 prices and quantities $(\$2 \times 80) + (\$2 \times 60) = \$280$, and the real GDP is calculated using Year 1 prices and Year 2 quantities $(\$1 \times 80) + (\$2 \times 60) = \$200$.
(D)	Incorrect. This answer incorrectly calculates the real GDP. Nominal GDP is calculated using Year 2 prices and quantities $(\$2 \times 80) + (\$2 \times 60) = \$280$, and the real GDP is calculated using Year 1 prices and Year 2 quantities $(\$1 \times 80) + (\$2 \times 60) = \$200$.
(E)	Incorrect. This answer incorrectly calculates the nominal and real GDP. Nominal GDP is calculated using Year 2 prices and quantities $(\$2 \times 80) + (\$2 \times 60) = \$280$, and the real GDP is calculated using Year 1 prices and Year 2 quantities $(\$1 \times 80) + (\$2 \times 60) = \$200$.

Question 9

Skill	Learning Objective	Topic
2.A	POL-1.A	Fiscal Policy

(A)	Incorrect. This action is a contractionary fiscal policy that will decrease aggregate demand and therefore will increase unemployment rather than reduce unemployment. An expansionary fiscal policy action, such as increasing government expenditures, decreasing taxes, or increasing transfer payments, would increase aggregate demand and reduce unemployment.
(B)	**Correct**. This action is an expansionary fiscal policy that will increase aggregate demand and therefore will decrease unemployment.
(C)	Incorrect. This action is a contractionary fiscal policy that will decrease aggregate demand and therefore will increase unemployment rather than reduce unemployment. An expansionary fiscal policy action, such as increasing government expenditures, decreasing taxes, or increasing transfer payments, would increase aggregate demand and reduce unemployment.
(D)	Incorrect. This action is not a fiscal policy. Government expenditures, taxes, and transfer payments are tools of fiscal policy.
(E)	Incorrect. This action is not a fiscal policy. Government expenditures, taxes, and transfer payments are tools of fiscal policy.

Question 10

Skill	Learning Objective	Topic
1.B	MEA-2.B	Economic Growth

(A)	Incorrect. Building more factories increases physical capital, not human capital.
(B)	Incorrect. Reducing immigration of skilled workers decreases human capital.
(C)	**Correct**. Improving the quality of job-training programs increases human capital.
(D)	Incorrect. Increasing the physical capital per worker increases the productivity of workers, not human capital.
(E)	Incorrect. Increasing government spending on infrastructure increases physical capital, not human capital.

Question 11

Skill	Learning Objective	Topic
3.C	MOD-2.B	Multipliers

(A)	**Correct**. National income will increase by a maximum of $\$60$ billion. The maximum change in national income is determined by multiplying the spending multiplier by the amount of the change in government spending. The spending multiplier is equal to $\left(\frac{1}{\text{MPS}} = \frac{1}{0.25} = 4\right)$. Therefore, national income will increase by a maximum of $\$15\,\text{billion} \times 4 = \$60\,\text{billion}$.
(B)	Incorrect. This answer mistakenly calculates the tax multiplier instead of the government spending multiplier. National income will increase by a maximum of $\$60$ billion. The maximum change in national income is determined by multiplying the spending multiplier by the amount of the change in government spending. The spending multiplier is equal to $\left(\frac{1}{\text{MPS}} = \frac{1}{0.25} = 4\right)$. Therefore, national income will increase by a maximum of $\$15\,\text{billion} \times 4 = \$60\,\text{billion}$.
(C)	Incorrect. This answer mistakenly calculates the government spending multiplier as 1. National income will increase by a maximum of $\$60$ billion. The maximum change in national income is determined by multiplying the spending multiplier by the amount of the change in government spending. The spending multiplier is equal to $\left(\frac{1}{\text{MPS}} = \frac{1}{0.25} = 4\right)$. Therefore, national income will increase by a maximum of $\$15\,\text{billion} \times 4 = \$60\,\text{billion}$.
(D)	Incorrect. This answer mistakenly assumes that national income will increase by the change in consumption as depicted by the marginal propensity to consume. National income will increase by a maximum of $\$60$ billion. The maximum change in national income is determined by multiplying the spending multiplier by the amount of the change in government spending. The spending multiplier is equal to $\left(\frac{1}{\text{MPS}} = \frac{1}{0.25} = 4\right)$. Therefore, national income will increase by a maximum of $\$15\,\text{billion} \times 4 = \$60\,\text{billion}$.
(E)	Incorrect. This answer mistakenly assumes that national income will increase by the change in savings as depicted by the marginal propensity to save. National income will increase by a maximum of $\$60$ billion. The maximum change in national income is determined by multiplying the spending multiplier by the amount of the change in government spending. The spending multiplier is equal to $\left(\frac{1}{\text{MPS}} = \frac{1}{0.25} = 4\right)$. Therefore, national income will increase by a maximum of $\$15\,\text{billion} \times 4 = \$60\,\text{billion}$.

Question 12

Skill	Learning Objective	Topic
3.B	POL-1.D	Monetary Policy

(A)	Incorrect. An increase in the money supply will increase aggregate demand and real output and decrease unemployment below the natural rate in the short run. This will create an inflationary gap because real output will exceed the full employment level of output, which will create a shortage of labor and overutilization of resources. In the long run, this will cause input prices and nominal wages to increase and shift the short-run aggregate supply curve to the left to restore full employment and increase (not decrease) unemployment back to the natural rate.
(B)	Incorrect. An increase in the money supply will increase aggregate demand and real output and decrease (not increase) unemployment below the natural rate in the short run. This will create an inflationary gap because real output will exceed the full employment level of output, which will create a shortage of labor and overutilization of resources. In the long run, this will cause input prices and nominal wages to increase and shift the short-run aggregate supply curve to the left to restore full employment and increase unemployment back to the natural rate.
(C)	Incorrect. An increase in the money supply will increase aggregate demand and real output, and unemployment will decrease below the natural rate (rather than remain unchanged) in the short run. This will create an inflationary gap because real output will exceed the full employment level of output, which will create a shortage of labor and overutilization of resources. In the long run, this will cause input prices and nominal wages to increase and shift the short-run aggregate supply curve to the left to restore full employment and increase unemployment back to the natural rate.
(D)	Incorrect. An increase in the money supply will increase aggregate demand and real output and decrease unemployment below the natural rate (not above the natural rate) in the short run. This will create an inflationary gap because real output will exceed the full employment level of output, which will create a shortage of labor and overutilization of resources. In the long run, this will cause input prices and nominal wages to increase and shift the short-run aggregate supply curve to the left to restore full employment and increase (not decrease) unemployment back to the natural rate.
(E)	**Correct**. An increase in the money supply will increase aggregate demand and real output and decrease unemployment below the natural rate in the short run. This will create an inflationary gap because real output will exceed the full employment level of output, which will create a shortage of labor and overutilization of resources. In the long run, this will cause input prices and nominal wages to increase and shift the short-run aggregate supply curve to the left to restore full employment and increase unemployment back to the natural rate.

Question 13

Skill	Learning Objective	Topic
1.A	MEA-1.A	The Circular Flow and GDP

(A)	Incorrect. The term "value added" is sales minus intermediate purchases, which are the costs of inputs purchased from other firms. The term "value added" also refers to the sum of the contributions of each firm.
(B)	Incorrect. The term "value added" is sales minus intermediate purchases, which are the costs of inputs purchased from other firms. The term "value added" also refers to the sum of the contributions of each firm.
(C)	Incorrect. The term "value added" is sales minus intermediate purchases, which are the costs of inputs purchased from other firms. The term "value added" also refers to the sum of the contributions of each firm.
(D)	**Correct**. The term "value added" is sales minus intermediate purchases, which are the costs of inputs purchased from other firms. The term "value added" also refers to the sum of the contributions of each firm.
(E)	Incorrect. The term "value added" is sales minus intermediate purchases, which are the costs of inputs purchased from other firms. The term "value added" also refers to the sum of the contributions of each firm.

Question 14

Skill	Learning Objective	Topic
3.A	MEA-4.A	Balance of Payments Accounts

(A)	Incorrect. The government budget is determined by government outlays and tax revenues. An increase in a country's current account surplus will not necessarily result in a decrease in the country's government budget surplus.
(B)	Incorrect. The country's national savings is determined by private savings and public savings. An increase in a country's current account surplus will not necessarily result in a decrease in the country's national savings.
(C)	Incorrect. An increase in a country's current account surplus will necessarily result in an increase (not a decrease) in the country's financial account deficit.
(D)	**Correct**. A current account surplus indicates that a nation is a net lender to the rest of the world. An increase in a country's current account surplus causes a country's financial capital outflows to increase.
(E)	Incorrect. An increase in national debt is determined by the government's budget deficit. An increase in a country's current account surplus will not necessarily result in an increase in the country's national debt.

Question 15

Skill	Learning Objective	Topic
2.C	MEA-1.F	Price Indices and Inflation

(A)	Incorrect. The GDP deflator is the ratio between nominal GDP and real GDP $\times 100$. Since the GDP deflator in 2011 was less than 100, 2011 prices were falling and therefore the inflation rate in 2011 was negative (not positive).
(B)	**Correct**. The GDP deflator is the ratio between nominal GDP and real GDP $\times 100$. Since the GDP deflator in 2011 was less than 100, 2011 prices were falling and therefore the inflation rate in 2011 was negative.
(C)	Incorrect. The GDP deflator is the ratio between nominal GDP and real GDP $\times 100$. Since the GDP deflator in 2011 was less than 100, 2011 prices were falling and therefore the inflation rate in 2011 was negative (not zero).
(D)	Incorrect. The GDP deflator is the ratio between nominal GDP and real GDP $\times 100$. Since the GDP deflator in 2011 was less than 100, 2011 prices were falling and therefore the purchasing power of a dollar increased (rather than decreased) by 3 percent.
(E)	Incorrect. The GDP deflator is a price index that measures changes in price level, not output.

Question 16

Skill	Learning Objective	Topic
2.A	MEA-2.A	Business Cycles

(A)	Incorrect. When the economy is in expansion, the actual real GDP is greater than the potential real GDP and the actual rate of unemployment is less than the natural rate of unemployment.
(B)	Incorrect. When the economy is in contraction, the actual real GDP is less than the potential real GDP and the actual rate of unemployment is greater than the natural rate of unemployment.
(C)	Incorrect. When the economy is at the peak, the actual real GDP is greater than the potential real GDP and the actual rate of unemployment is less than the natural rate of unemployment.
(D)	Incorrect. When the economy is at the trough, the actual real GDP is less than the potential real GDP and the actual rate of unemployment is greater than the natural rate of unemployment.
(E)	**Correct**. When the economy is on the potential line, the actual real GDP equals the potential real GDP and the actual rate of unemployment equals the natural rate of unemployment.

Question 17

Skill	Learning Objective	Topic
2.A	POL-1.A	Fiscal Policy

(A)	Incorrect. Increasing the required reserve ratio is a contractionary monetary policy action, not a fiscal policy action, and will decrease aggregate demand, real output, and the price level.
(B)	Incorrect. Increasing the discount rate is a contractionary monetary policy action, not a fiscal policy action, and will decrease aggregate demand, real output, and the price level.
(C)	Incorrect. Increasing transfer payments is an expansionary fiscal policy action that will increase aggregate demand, real output, and the price level, which will increase (not reduce) inflation.
(D)	Incorrect. Increasing government spending is an expansionary fiscal policy action that will increase aggregate demand, real output, and the price level, which will increase (not reduce) inflation.
(E)	**Correct**. Increasing income tax rates is a contractionary fiscal policy action that will decrease aggregate demand, real output, and the price level, which will decrease inflation.

Question 18

Skill	Learning Objective	Topic
1.A	MEA-1.A	The Circular Flow and GDP

(A)	Incorrect. Financial assets are not accounted for in GDP. Nominal GDP is a measure of the total value of final goods and services produced domestically.
(B)	Incorrect. The total value of firms located within the borders of China is not how nominal GDP is measured. Nominal GDP is a measure of the total value of final goods and services produced domestically.
(C)	Incorrect. Final goods and services consumed in China are counted as one component of GDP but do not represent the total value of GDP. Nominal GDP is a measure of the total value of final goods and services produced domestically.
(D)	**Correct**. Nominal GDP is a measure of the total value of final goods and services produced domestically.
(E)	Incorrect. Final goods and services exported by China to the rest of the world are counted as one component of GDP but do not represent the total value of GDP. Nominal GDP is a measure of the total value of final goods and services produced domestically.

Question 19

Skill	Learning Objective	Topic
3.A	POL-1.A	Fiscal Policy

(A)	Incorrect. A decrease in government borrowing decreases the demand for loanable funds, which decreases (not increases) real interest rates. An increase in income taxes increases public savings and the supply of loanable funds, which also decreases (not increases) real interest rates in the short run.
(B)	**Correct**. A decrease in government spending and an increase in income taxes are contractionary fiscal policy actions aimed at decreasing aggregate demand, real output, and the price level. A decrease in real output is accompanied by an increase in unemployment.
(C)	Incorrect. A decrease in government spending and an increase in income taxes are contractionary fiscal policy actions and have no effect on the money supply. The money supply is determined by the central bank.
(D)	Incorrect. A decrease in government spending and an increase in income taxes are contractionary fiscal policy actions aimed at decreasing aggregate demand, real output, and the price level.
(E)	Incorrect. A decrease in government borrowing decreases the demand for loanable funds, which decreases real interest rates, thereby disincentivizing (rather than increasing) personal saving. An increase in income taxes increases public (not private) savings and the supply of loanable funds, which also decreases real interest rates in the short run.

Question 20

Skill	Learning Objective	Topic
1.A	MEA-3.B	Nominal v. Real Interest Rates

(A)	Incorrect. The real interest rate is the nominal interest rate adjusted for inflation and is equal to the nominal interest rate minus the expected inflation rate. Therefore, the higher the inflation rate, the lower (not higher) the real interest rate.
(B)	**Correct**. The real interest rate is the nominal interest rate adjusted for inflation and is equal to the nominal interest rate minus the expected inflation rate. Therefore, if there is no actual or expected inflation, the nominal and real interest rates will be equal.
(C)	Incorrect. The real interest rate is the nominal interest rate adjusted for inflation and is equal to the nominal interest rate minus the expected inflation rate. Deflation is negative inflation. Therefore, if the economy is experiencing deflation, the nominal interest rate will be less than (not exceed) the real interest rate.
(D)	Incorrect. The nominal interest rate is established as the sum of the expected real interest rate and the expected inflation rate. Therefore, the higher the inflation rate, the higher (not lower) the nominal interest rate.
(E)	Incorrect. The nominal interest rate is established as the sum of the expected real interest rate and the expected inflation rate, not the difference between the real interest rate and the expected inflation rate.

Question 21

Skill	Learning Objective	Topic
1.A	MOD-1.C	Economic Growth

(A)	Incorrect. The aggregate demand curve shows the relationship between the price level and the amount of spending on goods and services in the economy. A rightward shift in the aggregate demand curve illustrates positive demand shocks, not economic growth.
(B)	Incorrect. The long-run Phillips curve shows the relationship between the natural rate of unemployment and the inflation rate in the long run. A leftward (not rightward) shift in the long-run Phillips curve illustrates economic growth.
(C)	**Correct**. The production possibilities curve shows the amount of goods and services the economy can produce given its resources and given the current state of technology. An increase in productivity or technology will increase economic growth and shift the production possibilities curve to the right. Therefore, a rightward shift in the production possibilities curve illustrates economic growth.
(D)	Incorrect. The short-run aggregate supply curve shows the relationship between the price level and the amount of goods and services that will be supplied in the economy in the short run. A rightward shift in the short-run aggregate supply curve illustrates positive supply shocks, not economic growth.
(E)	Incorrect. The money supply curve shows the relationship between interest rates and the quantity of money supplied in the economy. A rightward shift in the money supply curve illustrates expansionary monetary policy, not economic growth.

Question 22

Skill	Learning Objective	Topic
1.A	MOD-2.H	Changes in the AD–AS Model in the Short Run

(A)	Incorrect. Stagflation is a condition of economic stagnation with relatively high unemployment accompanied by high (not low) inflation. Stagflation is usually caused by a decrease in short-run aggregate supply.
(B)	Incorrect. Stagflation is a condition of economic stagnation with relatively high (not low) unemployment accompanied by high inflation. Stagflation is usually caused by a decrease in short-run aggregate supply.
(C)	Incorrect. Stagflation is a condition of economic stagnation with relatively high unemployment accompanied by high (not low) inflation. Stagflation is usually caused by a decrease in short-run aggregate supply.
(D)	**Correct**. Stagflation is a condition of economic stagnation with relatively high unemployment accompanied by high inflation. Stagflation is usually caused by a decrease in short-run aggregate supply.
(E)	Incorrect. Stagflation is a condition of economic stagnation with relatively high unemployment accompanied by high inflation. Stagflation is usually caused by a decrease in short-run aggregate supply, not a decrease in aggregate demand resulting from changes in taxes and government spending.

Question 23

Skill	Learning Objective	Topic
3.A	MKT-4.E	The Loanable Funds Market

(A)	Incorrect. If businesses become optimistic about the profitability of investments in an economy, investment demand will increase, which will increase the demand for loanable funds. The supply of loanable funds is determined by national savings and net financial capital flows, so it will not change.
(B)	Incorrect. If businesses become optimistic about the profitability of investments in an economy, investment demand will increase, which will increase (not decrease) the demand for loanable funds. The supply of loanable funds is determined by national savings and net financial capital flows, so it will not change.
(C)	Incorrect. If businesses become optimistic about the profitability of investments in an economy, investment demand will increase, which will increase (not decrease) the demand for loanable funds.
(D)	**Correct**. If businesses become optimistic about the profitability of investments in an economy, investment demand will increase, which will increase the demand for loanable funds. An increase in the demand for loanable funds will increase the real interest rate.
(E)	Incorrect. If businesses become optimistic about the profitability of investments in an economy, investment demand will increase, which will increase the demand for loanable funds. An increase in the demand for loanable funds will increase (not decrease) the real interest rate.

Question 24

Skill	Learning Objective	Topic
2.B	POL-3.C	Crowding Out

(A)	**Correct**. Crowding out refers to the adverse effect of increased government borrowing, which leads to decreased levels of interest-sensitive private sector spending in the short run. If investment demand becomes less responsive to changes in interest rates, there will be less crowding out.
(B)	Incorrect. Crowding out refers to the adverse effect of increased government borrowing, which leads to decreased levels of interest-sensitive private sector spending in the short run. If investment demand becomes less responsive to changes in interest rates, there will be less (not more) crowding out.
(C)	Incorrect. Monetary policy refers to the central bank's actions influencing interest rates to affect interest-sensitive spending. If investment demand becomes less responsive to changes in interest rates, then monetary policy will become less (not more) effective.
(D)	Incorrect. Monetary policy refers to the central bank's actions influencing interest rates to affect interest-sensitive spending. If investment demand becomes less responsive to changes in interest rates, then monetary policy will become less (not more) effective.
(E)	Incorrect. Crowding out occurs when interest rates increase, not decrease. An expansionary monetary policy would reduce interest rates, and therefore there would be no crowding out.

Question 25

Skill	Learning Objective	Topic
1.A	MEA-3.A	Financial Assets

(A)	Incorrect. Leisure is time spent not working. Therefore, the nominal wage is the opportunity cost of time spent on leisure (not the opportunity cost of holding currency).
(B)	Incorrect. An increase in the demand for money is not an opportunity cost of holding currency. Currency is one component of money.
(C)	**Correct**. The most likely alternative to holding currency would be to hold an interest-bearing financial asset. Therefore, the opportunity cost of holding currency is the interest foregone from not holding an interest-bearing asset.
(D)	Incorrect. The ability to meet unexpected expenses is a benefit (not an opportunity cost) of holding currency.
(E)	Incorrect. The percentage of total financial assets held in the form of currency is unrelated to income tax rates. Therefore, the income tax rates are not a cost of holding currency.

Question 26

Skill	Learning Objective	Topic
3.B	POL-1.F	Fiscal and Monetary Policy Actions in the Short Run

(A)	Incorrect. A contractionary monetary policy will decrease aggregate demand and real output, which is consistent with a decrease in income and consumption. An expansionary fiscal policy will increase aggregate demand and real output, which is consistent with an increase in income and consumption. Therefore, the net effect of a combined contractionary monetary policy and expansionary fiscal policy on income and consumption is indeterminate.
(B)	Incorrect. A contractionary monetary policy will decrease aggregate demand and real output, which is consistent with a decrease in income and consumption. An expansionary fiscal policy will increase aggregate demand and real output, which is consistent with an increase in income and consumption. Therefore, the net effect of a combined contractionary monetary policy and expansionary fiscal policy on income and consumption is indeterminate.
(C)	Incorrect. Both contractionary monetary policy and expansionary fiscal policy increase interest rates. An increase in interest rates will decrease investment.
(D)	**Correct**. A contractionary monetary policy will decrease the money supply, which will increase nominal interest rates. An expansionary fiscal policy will increase aggregate demand, which will increase real output and the price level; this will lead to an increase in the demand for money and higher nominal interest rates. Therefore, both contractionary monetary policy and expansionary fiscal policy will increase interest rates. An increase in interest rates will decrease interest-sensitive spending, including investment.
(E)	Incorrect. Both contractionary monetary policy and expansionary fiscal policy will increase interest rates. An increase in interest rates will decrease (not increase) investment.

Question 27

Skill	Learning Objective	Topic
3.C	POL-2.A	Banking and the Expansion of the Money Supply

(A)	Incorrect. Required reserves are equal to the reserve requirement times the amount of checkable deposits. Excess reserves are equal to total reserves minus required reserves. The bank can loan out excess reserves. Therefore, of the $1,000 deposit, the bank must keep 10% as required reserves, or $100. The bank can lend out the rest of the deposit, or $1,000 – $100 = $900.
(B)	Incorrect. Required reserves are equal to the reserve requirement times the amount of checkable deposits. Excess reserves are equal to total reserves minus required reserves. The bank can loan out excess reserves. Therefore, of the $1,000 deposit, the bank must keep 10% as required reserves, or $100. The bank can lend out the rest of the deposit, or $1,000 – $100 = $900.
(C)	Incorrect. Required reserves are equal to the reserve requirement times the amount of checkable deposits. Excess reserves are equal to total reserves minus required reserves. The bank can loan out excess reserves. Therefore, of the $1,000 deposit, the bank must keep 10% as required reserves, or $100. The bank can lend out the rest of the deposit, or $1,000 – $100 = $900.
(D)	**Correct**. Required reserves are equal to the reserve requirement times the amount of checkable deposits. Excess reserves are equal to total reserves minus required reserves. The bank is allowed to loan out excess reserves. Therefore, of the $1,000 deposit, the bank must keep 10% as required reserves, or $100. The bank can lend out the rest of the deposit, or $1,000 – $100 = $900.
(E)	Incorrect. Required reserves are equal to the reserve requirement times the amount of checkable deposits. Excess reserves are equal to total reserves minus required reserves. The bank can loan out excess reserves. Therefore, of the $1,000 deposit, the bank must keep 10% as required reserves, or $100. The bank can lend out the rest of the deposit, or $1,000 – $100 = $900.

Question 28

Skill	Learning Objective	Topic
2.A	MEA-4.A	Balance of Payments Accounts

(A)	Incorrect. An increase in domestic inflation would cause domestic goods and services to become more expensive relative to foreign goods and services and, therefore, would lower exports and raise imports, which, in turn, would decrease (not increase) a balance of trade surplus.
(B)	**Correct**. A country will have a surplus in the balance of trade if exports exceed imports. Declining imports and rising exports will increase the surplus in the balance of trade.
(C)	Incorrect. Higher tariffs imposed by a trading partner would decrease a country's exports and reduce (not increase) a balance of trade surplus.
(D)	Incorrect. An increase in financial capital inflows will increase the capital and financial account. An increase in the capital and financial account will be offset by a decrease in the current account. The balance of trade is one component of the current account. Therefore, an increased surplus in its balance of trade will be offset by a decrease (not an increase) in financial capital inflows.
(E)	Incorrect. An appreciation of a country's currency would cause goods and services in the domestic country to become more expensive relative to foreign goods and services, resulting in a decrease in the country's exports and an increase in its imports. Therefore, an appreciation of the currency would lead to a decrease in the country's net exports and would decrease (not increase) a balance of trade surplus.

Question 29

Skill	Learning Objective	Topic
2.C	POL-3.B	Government Deficits and the National Debt

(A)	Incorrect. This response will increase (not decrease) the government budget deficit. If government outlays fall by $100 million and tax revenues fall by $600 million, the government budget deficit will increase by $500 million.
(B)	Incorrect. This response will not affect the budget deficit since the $200 million fall in government outlays and $200 million fall in tax revenues offset each other.
(C)	Incorrect. This response will increase (not decrease) the government budget deficit. If government outlays rise by $300 million and tax revenues fall by $300 million, the government budget deficit will increase by $600 million.
(D)	**Correct**. If government outlays rise by $400 million and tax revenues rise by $600 million, the government budget deficit will decrease by $200.
(E)	Incorrect. This response will not affect the budget deficit since the $500 million rise in government outlays and $500 million rise in tax revenues offset each other.

Question 30

Skill	Learning Objective	Topic
2.C	MKT-1.A	Comparative Advantage and Gains from Trade

(A)	Incorrect. Country Y, not Country X, has an absolute and comparative advantage in the production of motorcycles. Country Y can produce more motorcycles than Country X when all resources are devoted to the production of motorcycles; therefore, Country Y has an absolute advantage in the production of motorcycles. Country Y's opportunity cost of producing one motorcycle is $\frac{8}{15}$ automobile (0.53 automobile), while Country X's opportunity cost of producing one motorcycle is $\frac{8}{5}$ automobiles (1.6 automobiles). Therefore, Country Y has a comparative advantage in the production of motorcycles because it has a lower opportunity cost in the production of motorcycles than Country X.
(B)	Incorrect. Country X does not have an absolute and comparative advantage in the production of both goods since Country Y has an absolute advantage in the production of motorcycles and a comparative advantage in the production of motorcycles. Country Y can produce more motorcycles than Country X when all resources are devoted to the production of motorcycles; therefore, it has an absolute advantage in the production of motorcycles. Country Y's opportunity cost of producing one motorcycle is $\frac{8}{15}$ automobile (0.53 automobile), while Country X's opportunity cost of producing one motorcycle is $\frac{8}{5}$ automobiles (1.6 automobiles). Therefore, Country Y has a comparative advantage in the production of motorcycles because it has a lower opportunity cost in the production of motorcycles than Country X.
(C)	Incorrect. Country Y's opportunity cost of producing one motorcycle is $\frac{8}{15}$ automobile (0.53 automobile), while Country X's opportunity cost of producing one motorcycle is $\frac{8}{5}$ automobiles (1.6 automobiles). Therefore, Country Y has a comparative advantage in the production of motorcycles because it has a lower opportunity cost in the production of motorcycles than Country X.

Question 30 (continued)

(D)	Incorrect. Country X, not Country Y, has an absolute and comparative advantage in the production of automobiles. Country X can produce more automobiles than Country Y when all resources are devoted to the production of automobiles; therefore, it has an absolute advantage in the production of automobiles. Country X's opportunity cost of producing one automobile is $\frac{5}{8}$ motorcycle (0.625 motorcycle), while Country Y's opportunity cost of producing one automobile is $\frac{15}{8}$ motorcycles (1.875 motorcycles). Therefore, Country X has a comparative advantage in the production of automobiles because it has a lower opportunity cost in the production of automobiles than Country Y.
(E)	**Correct**. Country Y can produce more motorcycles than Country X when all resources are devoted to the production of motorcycles; therefore, it has an absolute advantage in the production of motorcycles. Country Y's opportunity cost of producing one motorcycle is $\frac{8}{15}$ automobile (0.53 automobile), while Country X's opportunity cost of producing one motorcycle is $\frac{8}{5}$ automobiles (1.6 automobiles). Therefore, Country Y has a comparative advantage in the production of motorcycles because it has a lower opportunity cost in the production of motorcycles than Country X.

Question 31

Skill	Learning Objective	Topic
1.D	MEA-1.G	Price Indices and Inflation

(A)	Incorrect. Energy prices do not have a higher impact on inflation than other input costs do. The goods in the market basket are weighted by their prices.
(B)	Incorrect. New products are not overrepresented in the CPI; they may be underrepresented.
(C)	Incorrect. The CPI does not assign greater weight to measures of welfare than it does to economic activity.
(D)	**Correct**. Product improvements are not always fully reflected in the calculation of the CPI. This will result in an overstatement of the cost of living.
(E)	Incorrect. The CPI does not adjust for the substitution of less-expensive goods by consumers.

Question 32

Skill	Learning Objective	Topic
3.C	MEA-3.C	Definition, Measurement, and Functions of Money

(A)	Incorrect. M1 is composed of currency in circulation and demand deposits. M2 is composed of M1 and other short-term and long-term savings accounts. Therefore, withdrawing money from savings accounts will increase M1 by increasing the currency in circulation but will not affect M2.
(B)	**Correct**. M1 is composed of currency in circulation and demand deposits. M2 is composed of M1 and other short-term and long-term savings accounts. Therefore, withdrawing money from savings accounts will increase M1 by increasing the currency in circulation but will not affect M2.
(C)	Incorrect. M1 is composed of currency in circulation and demand deposits. M2 is composed of M1 and other short-term and long-term savings accounts. Therefore, withdrawing money from savings accounts will increase M1 by increasing the currency in circulation but will not affect M2.
(D)	Incorrect. M1 is composed of currency in circulation and demand deposits. M2 is composed of M1 and other short-term and long-term savings accounts. Therefore, withdrawing money from savings accounts will increase M1 by increasing the currency in circulation but will not affect M2.
(E)	Incorrect. M1 is composed of currency in circulation and demand deposits. M2 is composed of M1 and other short-term and long-term savings accounts. Therefore, withdrawing money from savings accounts will increase M1 by increasing the currency in circulation but will not affect M2.

Question 33

Skill	Learning Objective	Topic
1.A	MOD-2.I	Long-Run Self-Adjustment

(A)	Incorrect. When the actual inflation rate exceeds the expected inflation rate, the economy is operating above full employment and, therefore, the actual unemployment rate is less than the natural rate of unemployment.
(B)	**Correct**. When wages fully adjust to expected inflation, the short-run aggregate supply curve will shift to the left until real output equals full employment and, therefore, the actual unemployment rate equals the natural rate of unemployment.
(C)	Incorrect. Expectations are not always correct in the short run. Expected inflation can vary from actual inflation depending on changes in economic conditions and fiscal and monetary policies.
(D)	Incorrect. The actual inflation rate is not always equal to the expected inflation rate because of changes in economic conditions and fiscal and monetary policies.
(E)	Incorrect. When the actual inflation rate equals the expected inflation rate, the economy is operating at full employment and, therefore, the natural rate of unemployment is equal to the unemployment rate (not the inflation rate).

Question 34

Skill	Learning Objective	Topic
1.B	MEA-4.A	Balance of Payments Accounts

(A)	Incorrect. This transaction is included in Country X's current account under exports, not in the capital and financial account.
(B)	Incorrect. This transaction is included in Country X's current account under net unilateral transfers, not in the capital and financial account.
(C)	Incorrect. This transaction is included in Country X's current account under net income from abroad, not in the capital and financial account.
(D)	Incorrect. This transaction is included in Country X's current account under unilateral transfers, not in the capital and financial account.
(E)	**Correct**. The financial account includes the net acquisition and disposal of financial assets and liabilities. Buying foreign government bonds is a financial transaction that is included in the financial account.

Question 35

Skill	Learning Objective	Topic
3.B	POL-1.F	Fiscal and Monetary Policy Actions in the Short Run

(A)	Incorrect. An increase in taxes is a contractionary fiscal policy that will decrease (not increase) aggregate demand. Decreasing the money supply is a contractionary monetary policy that will decrease (not increase) aggregate demand. Both policies are contractionary and will increase unemployment in the short run.
(B)	Incorrect. An increase in taxes is a contractionary fiscal policy that will decrease aggregate demand (not increase aggregate supply). Decreasing the money supply is a contractionary monetary policy that will decrease aggregate demand (not increase aggregate supply). Both policies are contractionary and will increase unemployment in the short run.
(C)	Incorrect. An increase in taxes is a contractionary fiscal policy that will decrease consumption spending (not increase investment spending). Decreasing the money supply is a contractionary monetary policy that will increase interest rates, which will decrease interest-sensitive spending (not increase investment spending). Both policies are contractionary and will increase unemployment in the short run.
(D)	**Correct**. An increase in taxes is a contractionary fiscal policy that will decrease aggregate demand and real output. Decreasing the money supply is a contractionary monetary policy that will decrease aggregate demand and real output. Both policies are contractionary and will increase unemployment in the short run.
(E)	Incorrect. An increase in taxes is a contractionary fiscal policy that will decrease aggregate demand and the price level. Decreasing the money supply is a contractionary monetary policy that will decrease aggregate demand and the price level. Both policies will decrease the price level; therefore, inflation will not increase in the short run.

Question 36

Skill	Learning Objective	Topic
3.A	MKT-4.D	The Loanable Funds Market

(A)	Incorrect. When there is excess demand (a shortage) in the loanable funds market, it means that the supply of loanable funds (national savings) falls short of (not exceeds) the demand for loanable funds (investment spending). This will put upward pressure on the real interest rate, which will drive the real interest rate toward equilibrium. Therefore, real interest rates will increase.
(B)	Incorrect. The condition of the economy in the context of the $AD - AS$ model cannot be determined from the given information in the loanable funds market. Excess demand (a shortage) in the loanable funds market will put upward pressure on the real interest rate, which will drive the real interest rate toward equilibrium. Therefore, real interest rates will increase.
(C)	**Correct**. Excess demand (a shortage) in the loanable funds market will put upward pressure on the real interest rate, which will drive the real interest rate toward equilibrium. Therefore, real interest rates will increase.
(D)	Incorrect. The condition of the economy in the context of the $AD - AS$ model cannot be determined from the given information in the loanable funds market. Excess demand (a shortage) in the loanable funds market will put upward pressure on the real interest rate, which will drive the real interest rate toward equilibrium. Therefore, real interest rates will increase.
(E)	Incorrect. The money supply will not change. The monetary base is determined by a country's central bank, so money supply is independent of the changes in the loanable funds market.

Question 37

Skill	Learning Objective	Topic
1.A	MOD-2.C	Short-Run Aggregate Supply (SRAS)

(A)	Incorrect. The underlying assumption of an upward-sloping short-run aggregate supply curve is that wages and prices are sticky or fixed in the short run.
(B)	Incorrect. The underlying assumption of an upward-sloping short-run aggregate supply curve is that wages and prices are sticky or fixed in the short run.
(C)	Incorrect. The underlying assumption of an upward-sloping short-run aggregate supply curve is that wages and prices are sticky or fixed in the short run.
(D)	**Correct**. When wages are sticky, an increase in the price level will increase profits, to which firms respond by hiring more workers and increasing production.
(E)	Incorrect. The underlying assumption of an upward-sloping short-run aggregate supply curve is that wages and prices are sticky or fixed in the short run.

Question 38

Skill	Learning Objective	Topic
1.A	MOD-3.A	The Phillips Curve

(A)	Incorrect. The short-run Phillips curve (SRPC) implies there is a trade-off between inflation and unemployment. Moving along the SRPC, an increase in inflation is associated with a decrease in unemployment.
(B)	Incorrect. The short-run Phillips curve (SRPC) implies there is a trade-off between inflation and unemployment. Moving along the SRPC, an increase in inflation is associated with a decrease in unemployment.
(C)	**Correct**. The short-run Phillips curve (SRPC) implies there is a trade-off between inflation and unemployment. Moving along the SRPC, an increase in inflation is associated with a decrease in unemployment.
(D)	Incorrect. The short-run Phillips curve (SRPC) implies there is a trade-off between inflation and unemployment. Moving along the SRPC, an increase in inflation is associated with a decrease in unemployment.
(E)	Incorrect. The short-run Phillips curve (SRPC) implies there is a trade-off between inflation and unemployment. Moving along the SRPC, an increase in inflation is associated with a decrease in unemployment.

Question 39

Skill	Learning Objective	Topic
2.C	MEA-3.B	Nominal v. Real Interest Rates

(A)	Incorrect. The real interest rate is equal to the nominal interest rate minus the expected inflation rate. The nominal interest rate is the unadjusted, stated rate of interest charged by the bank, independent of any expected changes in the price level. Therefore, the decrease in the real interest rate could have been caused by an increase in the actual inflation rate.
(B)	Incorrect. The real interest rate is equal to the nominal interest rate minus the expected inflation rate. The nominal interest rate is the unadjusted, stated rate of interest charged by the bank, independent of any expected changes in the price level. Therefore, the decrease in the real interest rate could have been caused by an increase in the actual inflation rate.
(C)	Incorrect. A decrease in the money supply would result in an increase in the nominal interest rate on new loans but not on existing loans. The real interest rate is equal to the nominal interest rate minus the expected inflation rate. Therefore, the decrease in the real interest rate could have been caused by an increase in the actual inflation rate.
(D)	**Correct**. The real interest rate is equal to the nominal interest rate minus the expected inflation rate. If the actual inflation rate turned out to be greater than the expected rate, the real rate would decrease.
(E)	Incorrect. The real interest rate is equal to the nominal interest rate minus the expected inflation rate. If the actual inflation rate turned out to be less than the expected rate, the real rate would increase (not decrease).

Question 40

Skill	Learning Objective	Topic
1.A	MEA-1.F	Price Indices and Inflation

(A)	Incorrect. Deflation is negative inflation, which is associated with a continuous decline in the CPI (not an increase).
(B)	Incorrect. Stagflation is a phenomenon that would occur when an economy experiences both high inflation and unemployment. Stagflation usually occurs when the economy experiences a negative aggregate supply shock, resulting in an increase in the price level and a decrease in real gross domestic product.
(C)	**Correct**. Inflation occurs when the aggregate price level as measured by the CPI increases. The percentage change in the CPI measures the inflation rate from one period to another.
(D)	Incorrect. During a recession, both real gross domestic product and the price level fall. The price level decreases (not increases) during a recession.
(E)	Incorrect. Disinflation is a decrease in the inflation rate. Usually, such an objective is achieved through contractionary monetary policy.

Question 41

Skill	Learning Objective	Topic
2.A	MOD-2.H	Changes in the AD–AS Model in the Short Run

(A)	**Correct**. A decrease in labor productivity decreases the output per worker and raises unit costs of production. The increase in costs of production will decrease short-run aggregate supply, raising the price level and lowering real output. This is a negative supply shock and will lead to cost-push inflation.
(B)	Incorrect. A decrease in income tax rates will increase aggregate demand in the short run, raising the price level and real output. Since the increase in the price level is caused by an increase in aggregate demand, the resulting inflation is demand-pull (not cost-push) inflation.
(C)	Incorrect. A decrease in optimism about future economic activity will decrease aggregate demand, resulting in a decrease in the price level and real output. As a result, inflation will decrease (not increase).
(D)	Incorrect. An increase in government spending will increase aggregate demand, resulting in an increase in the price level and real output. Since the increase in the price level is caused by an increase in aggregated demand, the resulting inflation is demand-pull (not cost-push) inflation.
(E)	Incorrect. Discovery of new sources of energy will increase short-run aggregate supply, resulting in a decrease in the price level and an increase in real output. This is a positive aggregate supply shock. As a result, inflation will decrease (not increase).

Question 42

Skill	Learning Objective	Topic
1.D	MEA-1.D	Unemployment

(A)	Incorrect. Individuals working in the underground economy are NOT counted as employed in the calculation of the measured unemployment rate because they are not part of the labor force. To be included in the unemployment statistics, individuals must be counted as part of the labor force. The measured unemployment rate is the percent of the labor force that is unemployed.
(B)	Incorrect. Individuals working more than one job are counted only once in the labor force as employed regardless of how many jobs they have. Therefore, the unemployment rate is not understated.
(C)	Incorrect. Because they have given up actively looking for jobs, discouraged workers are not part of the labor force. To be included in the unemployment statistics, individuals must be counted as part of the labor force. The measured unemployment rate is the percent of the labor force that is unemployed.
(D)	**Correct**. Because they have given up actively looking for jobs, discouraged workers are not part of the labor force. To be included in the unemployment statistics, individuals must be counted as part of the labor force. By not including discouraged workers, the measured unemployment rate understates the level of unemployment.
(E)	Incorrect. The measured unemployment rate is often criticized for understating the level of joblessness because it does NOT count groups such as part-time workers.

Question 43

Skill	Learning Objective	Topic
3.A	MOD-2.I	Long-Run Self-Adjustment

(A)	Incorrect. The economy is experiencing an inflationary output gap in the short run. In the long run, nominal wages will rise, causing the short-run aggregate supply curve to shift to the left until the economy returns back to potential output but at a higher price level. Therefore, output would decrease (not increase) and the price level would increase (not decrease).
(B)	Incorrect. The economy is experiencing an inflationary output gap in the short run. In the long run, nominal wages will rise, causing the short-run aggregate supply curve to shift to the left until the economy returns back to potential output but at a higher price level. Therefore, output would decrease (not increase) and the price level would increase.
(C)	Incorrect. The economy is experiencing an inflationary output gap in the short run. In the long run, nominal wages will rise, causing the short-run aggregate supply curve to shift to the left until the economy returns back to potential output but at a higher price level. Therefore, output would decrease and the price level would increase (not decrease).
(D)	**Correct**. The economy is experiencing an inflationary output gap in the short run. In the long run, nominal wages will rise, causing the short-run aggregate supply curve to shift to the left until the economy returns back to potential output but at a higher price level. Therefore, output would decrease and the price level would increase.
(E)	Incorrect. The economy is experiencing an inflationary output gap in the short run. In the long run, nominal wages will rise, causing the short-run aggregate supply curve to shift to the left until the economy returns back to potential output but at a higher price level. Therefore, output would decrease (not remain unchanged) and the price level would increase (not remain unchanged).

Question 44

Skill	Learning Objective	Topic
1.C	MOD-1.B	Opportunity Cost and the Production Possibilities Curve (PPC)

(A)	Incorrect. Moving from point A to point B requires giving up 4 shirts (production of shirts decreases from 20 to 16 to allocate the resources to produce the first chair), and moving from point B to point C requires giving up 4 shirts (production of shirts decreases from 16 to 12 to allocate the resources to produce the second chair), and so on. Therefore, the opportunity cost of producing one chair is constant and equal to 4 shirts (not zero).
(B)	**Correct**. Moving from point A to point B requires giving up 4 shirts (production of shirts decreases from 20 to 16 to allocate the resources to produce the first chair), and moving from point B to point C requires giving up 4 shirts (production of shirts decreases from 16 to 12 to allocate the resources to produce the second chair), and so on. Therefore, the opportunity cost of producing one chair is constant and equal to 4 shirts.
(C)	Incorrect. Moving from point A to point B requires giving up 4 shirts (production of shirts decreases from 20 to 16 to allocate the resources to produce the first chair), and moving from point B to point C requires giving up 4 shirts (production of shirts decreases from 16 to 12 to allocate the resources to produce the second chair), and so on. Therefore, the opportunity cost of producing one chair is constant (not increasing) and equal to 4 shirts.
(D)	Incorrect. Moving from point A to point B requires giving up 4 shirts (production of shirts decreases from 20 to 16 to allocate the resources to produce the first chair), and moving from point B to point C requires giving up 4 shirts (production of shirts decreases from 16 to 12 to allocate the resources to produce the second chair), and so on. Therefore, the opportunity cost of producing one chair is constant (not decreasing) and equal to 4 shirts.
(E)	Incorrect. Moving from point A to point B requires giving up 4 shirts (production of shirts decreases from 20 to 16 to allocate the resources to produce the first chair), and moving from point B to point C requires giving up 4 shirts (production of shirts decreases from 16 to 12 to allocate the resources to produce the second chair), and so on. Therefore, the opportunity cost of producing one chair is constant (not indeterminate) and equal to 4 shirts.

Question 45

Skill	Learning Objective	Topic
2.A	MOD-2.H	Changes in the AD–AS Model in the Short Run

(A)	Incorrect. An increase in income tax rates decreases disposable income and causes a decrease (not an increase) in aggregate demand, resulting in a decrease in both real output and the price level (not an increase in the price level).
(B)	Incorrect. An increase in input prices raises the cost of production and causes a decrease in short-run aggregate supply, resulting in lower real output and a higher price level. The rising price level is caused by the negative aggregate supply shock (not by a positive aggregate demand shock).
(C)	**Correct**. An increase in government spending raises aggregate demand, increasing both real output and the price level. The increase in the price level is likely to result in inflation.
(D)	Incorrect. An increase in real interest rates will lead to a decrease (not increase) in aggregate demand as interest-sensitive consumption spending and private investment spending will decrease. This will cause a decrease in both the real output and the price level (not an increase in the price level).
(E)	Incorrect. An increase in savings implies that there will be a reduction in consumption spending. The decrease in consumption spending will cause aggregate demand to decrease (not increase) and the price level to fall (rather than rise).

Question 46

Skill	Learning Objective	Topic
2.A	POL-1.F	Fiscal and Monetary Policy Actions in the Short Run

(A)	Incorrect. Increasing government spending is a fiscal policy action and not a monetary policy action taken by the central bank.
(B)	Incorrect. Increasing income tax rates is a fiscal policy action and not a monetary policy action taken by the central bank.
(C)	Incorrect. A decrease in consumer spending decreases aggregate demand and the price level, lowering the demand for money and the interest rate. Therefore, in order to stabilize the interest rate, the central bank needs to take an action that will increase interest rates. Decreasing the required reserve ratio results in an increase in the money supply and a further decrease (not increase) in the interest rate.
(D)	Incorrect. A decrease in consumer spending decreases aggregate demand and the price level, lowering the demand for money and the interest rate. Therefore, in order to stabilize the interest rate, the central bank needs to take an action that will increase interest rates. Decreasing the discount rate results in an increase in the money supply and a further decrease (not increase) in the interest rate.
(E)	**Correct**. A decrease in consumer spending decreases aggregate demand and the price level, lowering the demand for money and the interest rate. Therefore, in order to stabilize the interest rate, the central bank needs to take an action that will increase interest rates. Selling government bonds decreases the money supply and increases the interest rate. The policy will offset the effect of the decrease in consumer spending on interest rates and stabilize the nominal interest rate.

Question 47

Skill	Learning Objective	Topic
2.A	MKT-4.E	The Loanable Funds Market

(A)	Incorrect. An increase in interest rates raises the cost of capital and therefore lowers the incentive to borrow funds for investment spending.
(B)	Incorrect. Inflation reduces the value of investment returns and therefore is not conducive to investment spending, which in turn reduces the incentive to invest in physical capital.
(C)	**Correct**. An increase in business confidence about future economic activity and profitability of investments induces investment spending. This expectation of future profitability will shift the demand curve for loanable funds to the right, as businesses will be willing to borrow more at each interest rate. As a result, the economy's physical capital stock will increase.
(D)	Incorrect. An increase in money demand results in an increase in nominal interest rates, which raises the cost of capital and therefore lowers the incentive to borrow funds for investment spending.
(E)	Incorrect. An increase in consumption spending at all income levels will result in a reduction in consumer savings, which decreases the supply of loanable funds, raising the real interest rate. An increase in the real interest rate decreases the incentive to invest in physical capital because it raises the cost of capital.

Question 48

Skill	Learning Objective	Topic
3.B	MKT-5.G	Real Interest Rates and International Capital Flows

(A)	Incorrect. An increase in government spending not financed by taxes implies that the government is borrowing to finance its spending, which will increase the demand for loanable funds and causes the real interest rate to increase. The increase in the real interest rate is likely to make the country's financial assets more attractive to foreign investors, and the value of the country's currency will appreciate, making its exports relatively expensive to other countries. Therefore, the country's net exports (exports minus imports) will decrease, not increase.
(B)	**Correct**. An increase in government spending not financed by taxes implies that the government is borrowing to finance its spending, which will increase the demand for loanable funds and causes the real interest rate to increase. The increase in the real interest rate is likely to make the country's financial assets more attractive to foreign investors, and the value of the country's currency will appreciate, making its exports relatively expensive to other countries. Therefore, the country's net exports (exports minus imports) will decrease.
(C)	Incorrect. An increase in government spending not financed by taxes implies that the government is borrowing to finance its spending, which will increase the demand for loanable funds and causes the real interest rate to increase. The increase in the real interest rate is likely to make the country's financial assets more attractive to foreign investors, and the value of the country's currency will appreciate, making its exports relatively expensive to other countries. Therefore, the country's net exports (exports minus imports) will decrease (not remain constant).
(D)	Incorrect. An increase in government spending not financed by taxes implies that the government is borrowing to finance its spending, which will increase the demand for loanable funds and causes the real interest rate to increase (not decrease). The increase in the real interest rate is likely to make the country's financial assets more attractive to foreign investors, and the value of the country's currency will appreciate, making its exports relatively expensive to other countries. Therefore, the country's net exports (exports minus imports) will decrease (not increase).
(E)	Incorrect. An increase in government spending not financed by taxes implies that the government is borrowing to finance its spending, which will increase the demand for loanable funds and causes the real interest rate to increase (not decrease). The increase in the real interest rate is likely to make the country's financial assets more attractive to foreign investors, and the value of the country's currency will appreciate, making its exports relatively expensive to other countries. Therefore, the country's net exports (exports minus imports) will decrease.

Question 49

Skill	Learning Objective	Topic
2.A	MKT-3.D	The Money Market

(A)	**Correct**. An increase in the price level will increase the demand for money because people will need more money to pay for goods and services. An increase in the demand for money shifts the demand curve to the right.
(B)	Incorrect. A decrease in the price level will decrease (not increase) the demand for money because people will need less money to pay for goods and services. A decrease in the demand for money shifts the demand curve to the left (not to the right).
(C)	Incorrect. An increase in interest rates increases the opportunity cost of holding money and decreases the quantity of money demanded. That is, it causes a movement upward along the given money demand curve, but it does not shift the curve.
(D)	Incorrect. A decrease in interest rates decreases the opportunity cost of holding money and increases the quantity of money demanded. That is, it causes a movement down along the given money demand curve, but it does not shift the curve.
(E)	Incorrect. A decrease in the nominal gross domestic product decreases the volume of total transactions in the economy; therefore, it causes the money demand curve to shift to the left, not to the right.

Question 50

Skill	Learning Objective	Topic
3.A	POL-3.A	Money Growth and Inflation

(A)	Incorrect. Changes in the money supply affect only nominal values in the long run but not real values. Assuming the economy is in long-run equilibrium when the money supply increases, in the long run, nominal wages will increase and return the economy back to full employment. There will be a temporary increase in output in the short run, but the economy will return back to full-employment output in the long run at a higher price level.
(B)	Incorrect. Changes in the money supply affect only nominal values in the long run but not real values. Assuming the economy is in long-run equilibrium when the money supply increased, in the long run, nominal wages will increase and return the economy back to full employment. There will be an increase in the price level in the short run, but the economy will return back to full-employment output in the long run at a higher price level.
(C)	**Correct**. Changes in the money supply affect only nominal values in the long run but not real values. Assuming the economy is in long-run equilibrium when the money supply increased, in the long run, nominal wages will increase and return the economy back to full employment. There will be an increase in the price level in the short run, but the economy will return back to full-employment output in the long run at a higher price level.
(D)	Incorrect. Changes in the money supply affect only nominal values in the long run but not real values. Assuming the economy is in long-run equilibrium when the money supply increased, in the long run, nominal wages will increase and return the economy back to full employment. There will be a temporary increase in output in the short run, but the economy will return back to full-employment output in the long run at a higher price level.
(E)	Incorrect. Changes in the money supply affect only nominal values in the long run but not real values. Assuming the economy is in long-run equilibrium when the money supply increased, in the long run, nominal wages will increase and return the economy back to full employment. There will be an increase in the price level in the short run, but the economy will return back to full-employment output in the long run at a higher price level.

Question 51

Skill	Learning Objective	Topic
1.B	MEA-1.E	Unemployment

(A)	Incorrect. The former mayor is not in the labor force and therefore not counted as unemployed.
(B)	Incorrect. Unemployment caused by a downturn in economic activity is cyclical, not frictional.
(C)	Incorrect. College students working part-time are not counted as unemployed.
(D)	**Correct**. A college graduate that is interviewing and actively looking for a job remains unemployed until making a decision to accept the job. This is considered frictionally unemployed.
(E)	Incorrect. The architect is considered structurally unemployed; since the job is permanently gone due to structural changes in the job market, the skills are no longer in demand.

Question 52

Skill	Learning Objective	Topic
3.A	MOD-2.I	Long-Run Self-Adjustment

(A)	Incorrect. Since the actual unemployment rate exceeds the natural rate of unemployment, the economy is experiencing a recessionary output gap. If there is no active policy action to correct the macroeconomic condition, in the long run nominal wages will fall, causing the short-run aggregate supply curve to shift to the right until the economy returns back to potential output at a lower price level. Therefore, prices will decrease (not increase).
(B)	Incorrect. Since the actual unemployment rate exceeds the natural rate of unemployment, the economy is experiencing a recessionary output gap. If there is no active policy action to correct the macroeconomic condition, in the long run nominal wages will fall, causing the short-run aggregate supply curve to shift to the right until the economy returns back to potential output. Therefore, output will increase and unemployment will decrease (not increase).
(C)	**Correct**. Since the actual unemployment rate exceeds the natural rate of unemployment, the economy is experiencing a recessionary output gap. If there is no active policy action to correct the macroeconomic condition, in the long run nominal wages will fall, causing the short-run aggregate supply curve to shift to the right until the economy returns back to potential output.
(D)	Incorrect. Since the actual unemployment rate exceeds the natural rate of unemployment, the economy is experiencing a recessionary output gap. If there is no active policy action to correct the macroeconomic condition, in the long run nominal wages will fall, causing the short-run aggregate supply curve (not the aggregate demand curve) to shift to the right until the economy returns back to potential output.
(E)	Incorrect. Long-run aggregate supply will not change, since none of the factors that determine it have changed. The long-run aggregate supply is determined by changes in the capital stock, labor force, technology, and productivity, not by changes in nominal values, such as nominal wages.

Question 53

Skill	Learning Objective	Topic
3.A	MKT-2.G	Market Equilibrium, Disequilibrium, and Changes in Equilibrium

(A)	Incorrect. A decrease in supply will result in an increase in price and a decrease in quantity. A decrease in demand will result in a decrease in price and a decrease in quantity. Therefore, the equilibrium price will be indeterminate (not decrease), while the equilibrium quantity will decrease.
(B)	Incorrect. A decrease in supply will result in an increase in price and a decrease in quantity. A decrease in demand will result in a decrease in price and a decrease in quantity. Therefore, the equilibrium price will be indeterminate (not decrease), while the equilibrium quantity will decrease (not increase).
(C)	Incorrect. A decrease in supply will result in an increase in price and a decrease in quantity. A decrease in demand will result in a decrease in price and a decrease in quantity. Therefore, the equilibrium price will be indeterminate (not increase), while the equilibrium quantity will decrease.
(D)	Incorrect. A decrease in supply will result in an increase in price and a decrease in quantity. A decrease in demand will result in a decrease in price and a decrease in quantity. Therefore, the equilibrium price will be indeterminate (not increase), while the equilibrium quantity will decrease (not be indeterminate).
(E)	**Correct**. A decrease in supply will result in an increase in price and a decrease in quantity. A decrease in demand will result in a decrease in price and a decrease in quantity. Therefore, the equilibrium price will be indeterminate, while the equilibrium quantity will decrease.

Question 54

Skill	Learning Objective	Topic
3.A	POL-1.A	Fiscal Policy

(A)	Incorrect. A balanced budget entails financing a given increase in government spending by an equal increase in lump-sum taxes. The balanced budget multiplier is equal to the government spending multiplier plus the tax multiplier, i.e. $\frac{1}{1-\text{mpc}}+\frac{-\text{mpc}}{1-\text{mpc}}=\frac{1-\text{mpc}}{1-\text{mpc}}=1$. Thus, the net impact of an equal increase in government spending and lump-sum taxes shifts the aggregate demand curve to the right by an amount equal to the initial increase in government spending. Autonomous changes in government spending and lump-sum taxes do not affect the SRAS curve. Therefore, the SRAS curve will not change.
(B)	Incorrect. A balanced budget entails financing a given increase in government spending by an equal increase in lump-sum taxes. The balanced budget multiplier is equal to the government spending multiplier plus the tax multiplier, i.e. $\frac{1}{1-\text{mpc}}+\frac{-\text{mpc}}{1-\text{mpc}}=\frac{1-\text{mpc}}{1-\text{mpc}}=1$. Thus, the net impact of an equal increase in government spending and lump-sum taxes shifts the aggregate demand curve to the right by an amount equal to the initial increase in government spending. Autonomous changes in government spending and lump-sum taxes do not affect the SRAS curve. Therefore, the SRAS curve will not change.
(C)	**Correct**. A balanced budget entails financing a given increase in government spending by an equal increase in lump-sum taxes. The balanced budget multiplier is equal to the government spending multiplier plus the tax multiplier, i.e. $\frac{1}{1-\text{mpc}}+\frac{-\text{mpc}}{1-\text{mpc}}=\frac{1-\text{mpc}}{1-\text{mpc}}=1$. Thus, the net impact of an equal increase in government spending and lump-sum taxes shifts the aggregate demand curve to the right by an amount equal to the initial increase in government spending. Autonomous changes in government spending and lump-sum taxes do not affect the SRAS curve. Therefore, the SRAS curve will not change.

Question 54 (continued)

(D)	Incorrect. A balanced budget entails financing a given increase in government spending by an equal increase in lump-sum taxes. The balanced budget multiplier is equal to the government spending multiplier plus the tax multiplier, i.e. $\frac{1}{1-\text{mpc}} + \frac{-\text{mpc}}{1-\text{mpc}} = \frac{1-\text{mpc}}{1-\text{mpc}} = 1.$ Thus, the net impact of an equal increase in government spending and lump-sum taxes shifts the aggregate demand curve to the right by an amount equal to the initial increase in government spending. Autonomous changes in government spending and lump-sum taxes do not affect the SRAS curve. Therefore, the SRAS curve will not change.
(E)	Incorrect. A balanced budget entails financing a given increase in government spending by an equal increase in lump-sum taxes. The balanced budget multiplier is equal to the government spending multiplier plus the tax multiplier, i.e. $\frac{1}{1-\text{mpc}} + \frac{-\text{mpc}}{1-\text{mpc}} = \frac{1-\text{mpc}}{1-\text{mpc}} = 1.$ Thus, the net impact of an equal increase in government spending and lump-sum taxes shifts the aggregate demand curve to the right by an amount equal to the initial increase in government spending. Autonomous changes in government spending and lump-sum taxes do not affect the SRAS curve. Therefore, the SRAS curve will not change.

Question 55

Skill	Learning Objective	Topic
3.B	MKT-5.F	Changes in the Foreign Exchange Market and Net Exports

(A)	Incorrect. The appreciating dollar will make United States goods and services relatively expensive to Japanese buyers, and Japanese demand for United States goods and services will decrease. United States exports to Japan will decrease (not increase), but United States imports from Japan will increase because of the appreciating dollar, resulting in an increase in the United States trade deficit with respect to Japan. This will lead to a deficit in the United States current account and a surplus (not a deficit) in the United States capital and financial account.
(B)	Incorrect. The appreciating dollar will make United States goods and services relatively expensive to Japanese buyers, and Japanese demand for United States goods and services will decrease. United States exports to Japan will decrease (not increase), but United States imports from Japan will increase because of the appreciating dollar, resulting in an increase in the United States trade deficit with respect to Japan. This will lead to a deficit in the United States current account and a surplus in the United States capital and financial account.
(C)	**Correct**. The appreciating dollar will make United States goods and services relatively expensive to Japanese buyers, and Japanese demand for United States goods and services will decrease. United States exports to Japan will decrease, but United States imports from Japan will increase because of the appreciating dollar, resulting in an increase in the United States trade deficit with respect to Japan. This will lead to a deficit in the United States current account and a surplus in the United States capital and financial account.
(D)	Incorrect. The appreciating dollar will make United States goods and services relatively expensive to Japanese buyers, and Japanese demand for United States goods and services will decrease. United States exports to Japan will decrease, but United States imports from Japan will increase because of the appreciating dollar, resulting in an increase in the United States trade deficit with respect to Japan. This will lead to a deficit in the United States current account and a surplus (not a deficit) in the United States capital and financial account.
(E)	Incorrect. The appreciating dollar will make United States goods and services relatively expensive to Japanese buyers, and Japanese demand for United States goods and services will decrease. United States exports to Japan will decrease (not remain unchanged), but United States imports from Japan will increase because of the appreciating dollar, resulting in an increase in the United States trade deficit with respect to Japan. This will lead to a deficit (not a surplus) in the United States current account and a surplus (not a deficit) in the United States capital and financial account.

Question 56

Skill	Learning Objective	Topic
3.B	POL-1.F	Fiscal and Monetary Policy Actions in the Short Run

(A)	Incorrect. The decrease in government spending will decrease aggregate demand, whereas the central bank's open-market bond purchase will increase aggregate demand. Since the relative magnitude of the two policies is unknown, the net impact on the aggregate demand curve is indeterminate.
(B)	Incorrect. The decrease in government spending will decrease aggregate demand, whereas the central bank's open-market bond purchase will increase aggregate demand. Since the relative magnitude of the two policies is unknown, the net impact on the aggregate demand curve is indeterminate.
(C)	Incorrect. Both policies affect only the aggregate demand curve, not the short-run aggregate supply curve.
(D)	**Correct**. The decrease in government spending decreases aggregate demand, which decreases real output and the price level, resulting in a decrease in the demand for money and a decrease in the interest rate. The open-market bond purchase by the central bank increases the money supply and decreases the interest rate. Therefore, both actions lead to a reduction in interest rates.
(E)	Incorrect. The decrease in government spending decreases aggregate demand, which decreases real output and the price level, resulting in a decrease in the demand for loanable funds and a decrease in the interest rate. The open-market bond purchase by the central bank increases the money supply and decreases the interest rate. Therefore, both actions lead to a reduction (not a rise) in interest rates.

Question 57

Skill	Learning Objective	Topic
1.B	MEA-2.B	Economic Growth

(A)	Incorrect. Productivity is measured in terms of output per worker. An expansion of the labor force will lead to an increase in total output, but it is not clear what will happen to output per worker.
(B)	Incorrect. Productivity is measured in terms of output per worker. An increase in the value of financial capital is not an indicator of a change in output per worker.
(C)	Incorrect. Productivity is measured in terms of output per worker. A decrease in the amount of physical capital per worker decreases the output per worker, so productivity decreases, not increases.
(D)	**Correct**. Productivity rises when output per worker rises or when there is a decrease in the amount of labor needed to produce a unit of output.
(E)	Incorrect. An increase in the amount of resources required to produce a certain level of output leads to a decrease (not an increase) in productivity.

Question 58

Skill	Learning Objective	Topic
2.A	MKT-5.E	Effect of Changes in Policies and Economic Conditions on the Foreign Exchange Market

(A)	**Correct**. An increase in the price level in a country will make the country's goods and services relatively more expensive to other countries, which decreases foreign demand for the country's goods and services. This decrease in demand causes a decrease in the demand for the country's currency, resulting in a depreciation of the country's currency.
(B)	Incorrect. An increase in the real interest rate will make financial investment in the country's financial assets more attractive to other countries and increase the demand for the country's currency, which will cause the country's currency to appreciate (not depreciate).
(C)	Incorrect. A decrease in the country's expected inflation will make the country's goods relatively cheaper to other countries, which increases the country's exports. This will increase the demand for the country's currency and cause the country's currency to appreciate (not depreciate).
(D)	Incorrect. A decrease in the country's real gross domestic product decreases the demand for imports, which decreases the supply of the country's currency in foreign exchange markets, resulting in an appreciation of the country's currency (not a depreciation).
(E)	Incorrect. A decrease in the country's money supply raises the interest rate, which makes financial investment in the country's financial assets more attractive to other countries and increases the demand for the country's currency, which causes the country's currency to appreciate (not depreciate).

Question 59

Skill	Learning Objective	Topic
1.A	POL-2.A	Banking and the Expansion of the Money Supply

(A)	**Correct**. Banks use a portion of customers' deposits to expand credit. Specifically, banks use their excess reserves, reserves in excess of the required reserves they are legally required to keep to meet deposit liabilities, to make loans. Thus banks create or expand the money supply when they make loans.
(B)	Incorrect. Banks use a portion of customers' deposits to expand credit. Specifically, banks use their excess reserves, reserves in excess of the required reserves they are legally required to keep to meet deposit liabilities, to make loans. Thus banks create or expand the money supply when they make loans, not when loans are repaid.
(C)	Incorrect. If banks keep all their excess reserves, then they are not making any loans and will not be able to create money and expand the money supply.
(D)	Incorrect. When people withdraw funds from their checking or savings deposits, it decreases bank reserves and reduces banks' ability to expand the money supply. That is, it reduces the amount of loans banks can make.
(E)	Incorrect. The money multiplier affects only the magnitude of the expansion of the money supply, not banks' ability to create money.

Question 60

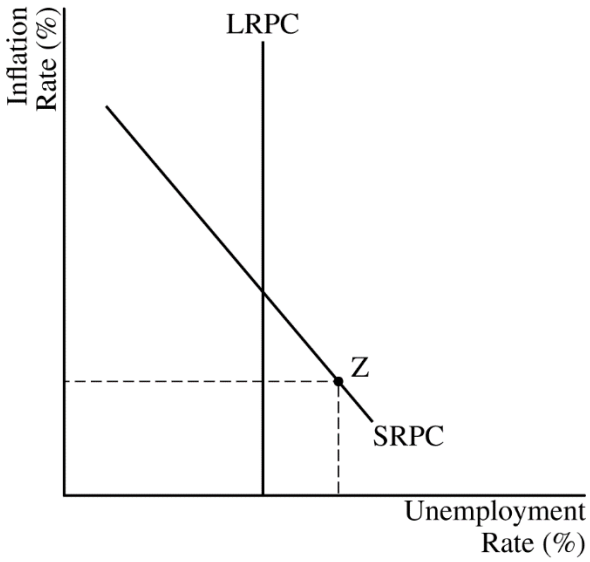
Skill	Learning Objective	Topic
1.A	POL-3.B	Government Deficits and the National Debt

(A)	Incorrect. A trade deficit occurs when a country's imports of goods and services exceed its exports. However, a country's national debt is not affected by its trade deficit or surplus.
(B)	Incorrect. The national debt increases when the country's government has a budget deficit. An increase in gross domestic product may improve a country's budget balance (rather than moving it into deficit and increasing the national debt) because tax revenues increase as GDP rises.
(C)	**Correct**. Each fiscal year, when a country's government spends more than it collects in tax revenues, it runs a budget deficit. The government often borrows funds to finance its deficit. The national debt grows as the government runs deficits year over year.
(D)	Incorrect. A country's national debt is not affected by its trade deficit or surplus.
(E)	Incorrect. National savings are the sum of private savings and government savings. When the government runs a budget deficit, national savings decrease. The national debt is not affected by the decrease in national savings. When private savings are not sufficient to finance current budget deficits, there will be a net financial capital inflow as the country borrows foreign savings to finance its deficits.

Question 1

10 points (2, 3, 2, 1, 1, 1)

(a) 2 points



- One point is earned for drawing a correctly labeled graph of the SRPC.
- One point is earned for drawing the LRPC and for showing point Z on the SRPC to the right of the LRPC.

b) 3 points:

- One point is earned for stating that the monetary base will change by $100 billion.
- One point is earned for correctly calculating the change in the amount of loans in the banking system as $300 (=4×$75).
- One point is earned for correctly calculating the change in the money supply as $400 (=4×$100).

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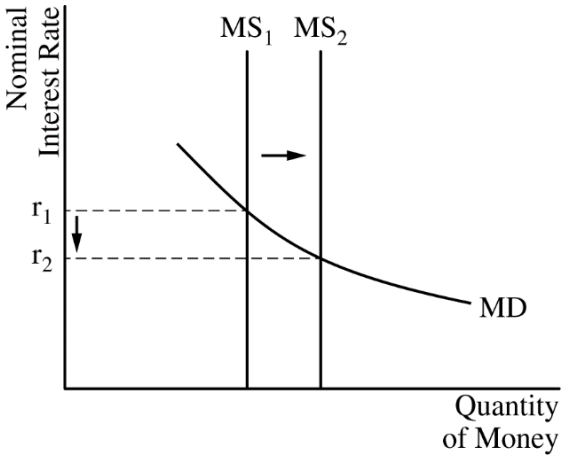
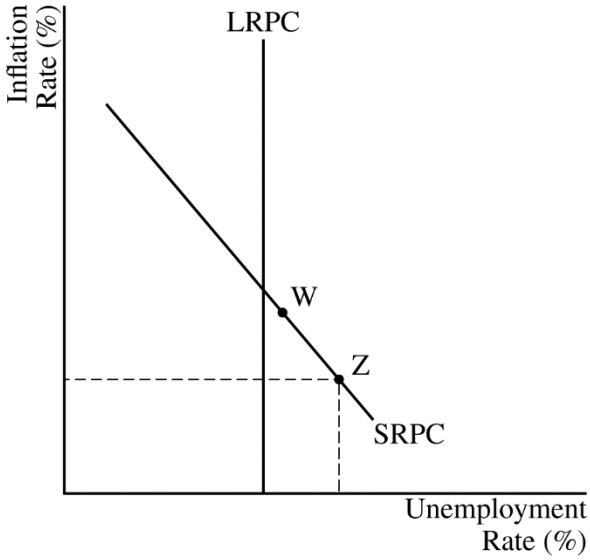
Question 1 (continued)

(c) 2 points



- One point is earned for drawing a correctly labeled graph of the money market.
- One point is earned for showing a rightward shift in the money supply curve and a lower nominal interest rate.

(d) 1 point



- One point is earned for showing point W to the left of point Z on the SRPC.

(e) 1 point
- One point is earned for stating that the country's currency will depreciate.

(f) 1 point
- One point is earned for stating that net exports will increase and for explaining that this is because the country's goods will be relatively cheaper than foreign goods.

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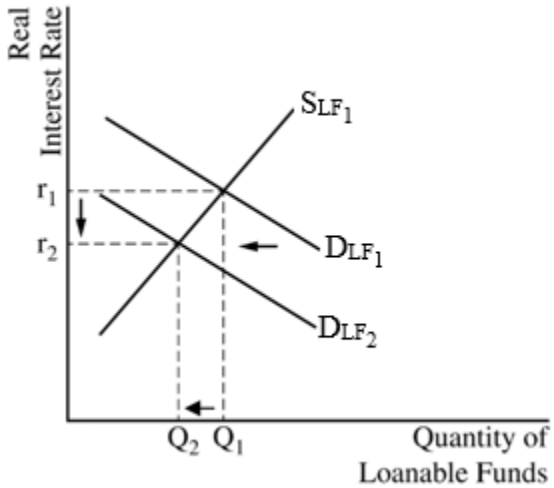
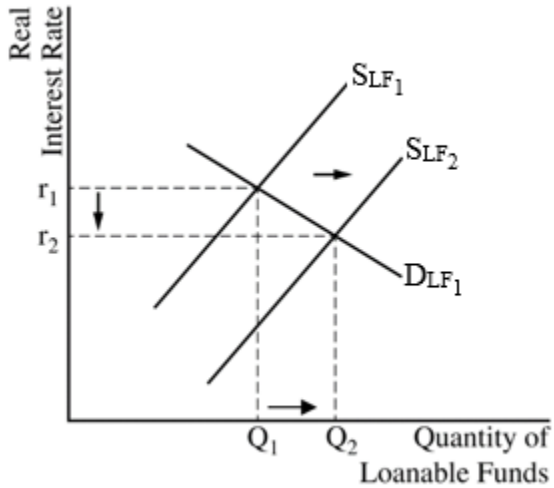
Question 2

5 Points (2, 1, 1, 1)

(a) 2 Points



OR



- One point is earned for drawing a correctly labeled graph of the loanable funds market.
- One point is earned for showing a leftward shift of the demand curve for loanable funds (or a rightward shift of the supply curve of loanable funds) and a decrease in the equilibrium real interest rate.

(b) 1 point

- One point is earned for stating that the dollar will depreciate against the euro and for explaining that the demand for United States financial assets decreases which decreases the demand for the dollar or increases the supply of the dollar. (It is also acceptable to explain that as investors seek higher returns in foreign financial assets, there will be financial capital outflows, which decreases the demand for the dollar or increases the supply of the dollar.)

(c) 1 point

- One point is earned for stating that the Federal Reserve should sell the euro and buy the dollar.

(d) 1 point

- One point is earned for stating that the Federal Reserve would sell bonds.

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Question 3

5 points (1, 1, 1, 1, 1)

(a) 1 point
- One point is earned for correctly calculating the nominal GDP for year 2 as 160 (=6×10+5×20).

(b) 1 point
- One point is earned for correctly calculating the real GDP for year 2 as 240 (=4×10+10×20).

(c) 1 point
- One point is earned for correctly calculating the GDP deflator for year 2 as 67 (=160/240 ×100).

(d) 1 point
- One point is earned for correctly calculating the CPI for year 2 as 110 (=110/100 ×100).

(e) 1 point
- One point is earned for correctly stating that real wages will decrease and for explaining that the increase in the nominal wage rate is lower than the inflation rate (2% < 10%).

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2019 AP Macroeconomics Scoring Worksheet

Section I: Multiple Choice

_____	× 1.0000 =	_____
Number Correct (out of 60)		Weighted Section I Score (Do not round)

Section II: Free Response

Question 1	_____ (out of 10)	× 1.5000 =	_____ (Do not round)
Question 2	_____ (out of 5)	× 1.5000 =	_____ (Do not round)
Question 3	_____ (out of 5)	× 1.5000 =	_____ (Do not round)

Sum = _____ Weighted Section II Score (Do not round)

Composite Score

_____	+	_____	=	_____
Weighted Section I Score		Weighted Section II Score		Composite Score (Round to nearest whole number)

AP Score Conversion Chart
Macroeconomics

Composite Score Range	AP Score
67-90	5
54-66	4
45-53	3
35-44	2
0-34	1

2019 AP Macroeconomics
Question Descriptors and Performance Data

Multiple-Choice Questions

Question	Skill	Learning Objective	Topic	Key	% Correct
1	2.A	MOD-1.B	Opportunity Cost and the Production Possibilities Curve (PPC)	D	53
2	2.A	MEA-2.B	Economic Growth	C	70
3	2.A	MOD-2.C	Short-Run Aggregate Supply (SRAS)	B	75
4	2.C	MKT-1.A	Comparative Advantage and Gains from Trade	C	72
5	3.A	MOD-2.C	Short-Run Aggregate Supply (SRAS)	B	84
6	2.A	MOD-2.H	Changes in the AD–AS Model in the Short Run	A	80
7	3.A	MOD-2.A	Aggregate Demand (AD)	B	58
8	1.C	MEA-1.J	Real v. Nominal GDP	C	68
9	2.A	POL-1.A	Fiscal Policy	B	76
10	1.B	MEA-2.B	Economic Growth	C	70
11	3.C	MOD-2.B	Multipliers	A	60
12	3.B	POL-1.D	Monetary Policy	E	52
13	1.A	MEA-1.A	The Circular Flow and GDP	D	26
14	3.A	MEA-4.A	Balance of Payments Accounts	D	33
15	2.C	MEA-1.F	Price Indices and Inflation	B	45
16	2.A	MEA-2.A	Business Cycles	E	76
17	2.A	POL-1.A	Fiscal Policy	E	63
18	1.A	MEA-1.A	The Circular Flow and GDP	D	74
19	3.A	POL-1.A	Fiscal Policy	B	59
20	1.A	MEA-3.B	Nominal v. Real Interest Rates	B	39
21	1.A	MOD-1.C	Economic Growth	C	61
22	1.A	MOD-2.H	Changes in the AD–AS Model in the Short Run	D	48
23	3.A	MKT-4.E	The Loanable Funds Market	D	43
24	2.B	POL-3.C	Crowding Out	A	32
25	1.A	MEA-3.A	Financial Assets	C	59
26	3.B	POL-1.F	Fiscal and Monetary Policy Actions in the Short Run	D	34
27	3.C	POL-2.A	Banking and the Expansion of the Money Supply	D	80
28	2.A	MEA-4.A	Balance of Payments Accounts	B	62
29	2.C	POL-3.B	Government Deficits and the National Debt	D	62
30	2.C	MKT-1.A	Comparative Advantage and Gains from Trade	E	80
31	1.D	MEA-1.G	Price Indices and Inflation	D	53
32	3.C	MEA-3.C	Definition, Measurement, and Functions of Money	B	27
33	1.A	MOD-2.I	Long-Run Self-Adjustment	B	56
34	1.B	MEA-4.A	Balance of Payments Accounts	E	35
35	3.B	POL-1.F	Fiscal and Monetary Policy Actions in the Short Run	D	64
36	3.A	MKT-4.D	The Loanable Funds Market	C	66
37	1.A	MOD-2.C	Short-Run Aggregate Supply (SRAS)	D	47

2019 AP Macroeconomics
Question Descriptors and Performance Data

Question	Skill	Learning Objective	Topic	Key	% Correct
38	1.A	MOD-3.A	The Phillips Curve	C	85
39	2.C	MEA-3.B	Nominal v. Real Interest Rates	D	27
40	1.A	MEA-1.F	Price Indices and Inflation	C	79
41	2.A	MOD-2.H	Changes in the AD–AS Model in the Short Run	A	42
42	1.D	MEA-1.D	Unemployment	D	69
43	3.A	MOD-2.I	Long-Run Self-Adjustment	D	46
44	1.C	MOD-1.B	Opportunity Cost and the Production Possibilities Curve (PPC)	B	68
45	2.A	MOD-2.H	Changes in the AD–AS Model in the Short Run	C	77
46	2.A	POL-1.F	Fiscal and Monetary Policy Actions in the Short Run	E	32
47	2.A	MKT-4.E	The Loanable Funds Market	C	60
48	3.B	MKT-5.G	Real Interest Rates and International Capital Flows	B	39
49	2.A	MKT-3.D	The Money Market	A	39
50	3.A	POL-3.A	Money Growth and Inflation	C	41
51	1.B	MEA-1.E	Unemployment	D	62
52	3.A	MOD-2.I	Long-Run Self-Adjustment	C	38
53	3.A	MKT-2.G	Market Equilibrium, Disequilibrium, and Changes in Equilibrium	E	56
54	3.A	POL-1.A	Fiscal Policy	C	44
55	3.B	MKT-5.F	Changes in the Foreign Exchange Market and Net Exports	C	31
56	3.B	POL-1.F	Fiscal and Monetary Policy Actions in the Short Run	D	42
57	1.B	MEA-2.B	Economic Growth	D	70
58	2.A	MKT-5.E	Effect of Changes in Policies and Economic Conditions on the Foreign Exchange Market	A	43
59	1.A	POL-2.A	Banking and the Expansion of the Money Supply	A	57
60	1.A	POL-3.B	Government Deficits and the National Debt	C	50

Free-Response Questions

Question	Skill	Learning Objective	Topic	Mean Score
1	3.A	MOD-3.A	The Phillips Curve	4.44
2	2.A	MKT-4.C	The Loanable Funds Market	2.25
3	1.C	MEA-1.I	Real v. Nominal GDP	1.86